

A Survey on Knowledge-Oriented Retrieval-Augmented Generation

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Retrieval-Augmented Generation (RAG) has gained significant attention in recent years for its potential to enhance natural language understanding and generation by combining large-scale retrieval systems with generative models. RAG leverages external knowledge sources, such as documents, databases, or structured data, to improve model performance and generate more accurate and contextually relevant outputs. This survey aims to provide a comprehensive overview of RAG by examining its fundamental components, including retrieval mechanisms, generation processes, and the integration between the two. We discuss the key characteristics of RAG, such as its ability to augment generative models with dynamic external knowledge, and the challenges associated with aligning retrieved information with generative objectives. We also present a taxonomy that categorizes RAG methods, ranging from basic retrieval-augmented approaches to more advanced models incorporating multi-modal data and reasoning capabilities. Additionally, we review the evaluation benchmarks and datasets commonly used to assess RAG systems, along with a detailed exploration of its applications in fields such as question answering, summarization, and information retrieval. Finally, we highlight emerging research directions and opportunities for improving RAG systems, such as enhanced retrieval efficiency, model interpretability, and domain-specific adaptations. This paper concludes by outlining the prospects for RAG in addressing real-world challenges and its potential to drive further advancements in natural language processing¹.

Additional Key Words and Phrases: Information Retrieval, Retrieval-augmented Generation, Large Language Model

ACM Reference Format:

Mingyue Cheng, Yucong Luo, Jie Ouyang, Qi Liu*, Huijie Liu, Li Li, Shuo Yu, Bohou Zhang, Jiawei Cao, Jie Ma, Daoyu Wang, and Enhong Chen. 2018. A Survey on Knowledge-Oriented Retrieval-Augmented Generation. In . ACM, New York, NY, USA, 50 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 INTRODUCTION

Retrieval-Augmented Generation (RAG)[127] has emerged as a key approach that integrates information retrieval with generative models to enhance natural language processing tasks. By leveraging external knowledge sources, RAG systems can generate more accurate and contextually relevant outputs, addressing complex challenges in areas like question answering[119], summarization[85], and open-domain dialogue. In recent years, a variety of RAG methods have been proposed, ranging from basic retrieval-augmented models to more advanced architectures incorporating multi-hop[190] reasoning and memory-augmented techniques[67]. These developments have highlighted the potential of RAG to improve the performance of NLP systems by dynamically combining retrieval and generation in a unified framework.

¹<https://github.com/USTCAGI/Awesome-Papers-Retrieval-Augmented-Generation>

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RAG models augment traditional language models by incorporating external knowledge sources, such as documents, databases, or structured data[93, 103], during the generation process. Unlike conventional models that rely solely on pre-trained parameters, RAG systems dynamically retrieve relevant information at generation time, allowing them to produce more informed and contextually accurate outputs. This approach addresses key limitations of traditional language models, such as their inability to access real-time or domain-specific knowledge, and mitigates the challenge of handling out-of-vocabulary or rare entities. For example, in question answering tasks[62, 192], RAG models retrieve relevant passages from large corpora to generate more precise and informative answers, while in summarization[85, 172], they leverage external documents to provide richer and more comprehensive summaries. Early successes in RAG have demonstrated significant improvements in a range of NLP applications, including open-domain question answering, where RAG systems have outperformed traditional generative models by incorporating relevant external context, and document-based summarization, where they produce summaries that better reflect the nuances of the source material.

At the core of RAG lies a knowledge-centric approach, which places external knowledge as a key factor in improving language generation. By incorporating relevant, real-time, and structured information, RAG models can significantly enhance their ability to generate contextually accurate and factually grounded content. This knowledge-centric perspective addresses one of the critical limitations of traditional language models, which are constrained by their training data and lack access to dynamic or domain-specific knowledge. The integration of external knowledge allows RAG models to not only retrieve and incorporate relevant details but also reason over multiple pieces of information, resulting in more nuanced and informed outputs. This shift towards knowledge augmentation enables models to perform more complex tasks, such as handling specialized topics[243, 245, 274], improving response relevance in dialogue systems, and generating high-quality summaries[85, 172] that reflect the true essence of source materials. By embedding knowledge into the generative process[50, 111], RAG opens new possibilities for applications that demand a high level of accuracy and contextual awareness.

Despite the promising advancements in Retrieval-Augmented Generation, several challenges remain that hinder the full potential of these models. One of the primary issues is knowledge selection, where the model must effectively identify the most relevant pieces of information from vast external sources. This task is particularly challenging given the large, noisy, and diverse nature of real-world knowledge corpora. Another critical challenge is knowledge retrieval, which involves retrieving the right information at generation time while balancing efficiency and relevance. Current retrieval techniques often struggle with accurately matching queries to knowledge sources, especially in highly specialized or ambiguous contexts. Additionally, knowledge-enhanced in-context reasoning presents a significant hurdle, as models must not only retrieve relevant knowledge but also effectively integrate and reason over it in a coherent and contextually appropriate manner. These challenges highlight the need for a deeper understanding and improved methodologies in the RAG space, making a comprehensive survey necessary to address both the current limitations and future opportunities in this rapidly evolving field.

The primary objective of this survey is to provide a comprehensive overview of RAG from a knowledge-centric perspective. By focusing on the integration of external knowledge sources into the generative process, this survey aims to review the recent advancements in RAG models, methods, and techniques. We will cover key models that have emerged, along with the fundamental approaches used in retrieval and generation, offering insights into how these methods address the unique challenges of knowledge selection, retrieval, and in-context reasoning. Additionally, we seek to highlight emerging trends and identify gaps in the existing research, particularly areas that require further exploration, such as multi-modal knowledge integration[16, 25, 195] and domain-specific applications[146, 168, 216, 267]. This survey's unique contribution lies in its focus on RAG from a knowledge-centric viewpoint, providing a unified

framework for understanding and advancing the field. The intended audience includes researchers, practitioners, and industry professionals who are interested in exploring how RAG can be leveraged for a variety of natural language processing tasks. As the field continues to evolve rapidly, this survey will serve as both a foundational resource and a guide to future research, offering insights into ongoing challenges and opportunities for innovation.

This survey is organized as follows to provide a structured and comprehensive exploration of Retrieval-Augmented Generation (RAG). In Section 2: Background, we begin by reviewing the foundational concepts of RAG and the key challenges faced by current models, setting the stage for understanding the significance of integrating external knowledge into language generation. Section 3: Overview of Retrieval-Augmented Generation delves deeper into the core principles of RAG, detailing the processes of knowledge retrieval, integration, and generation. Section 4: Characteristics and Challenges of RAG expands on the challenges identified earlier, such as knowledge selection, retrieval efficiency, and in-context reasoning, providing a thorough discussion of the complexities involved. Section 5: Basic and Advanced RAG Approaches outlines the range of methods used in RAG, from traditional retrieval-based models to more sophisticated, multi-modal approaches that incorporate additional reasoning and memory components. Section 6: RAG Evaluation [61, 188, 208] focuses on the benchmarks and datasets commonly used to assess RAG systems, offering a critical review of existing evaluation practices and metrics. Section 7: RAG Applications highlights key use cases and real-world applications where RAG has demonstrated improvements, followed by Section 8: Prospects and Future Directions, which discusses emerging trends and potential areas for future research. Finally, the survey concludes in Section 9, summarizing the findings and offering a perspective on the future impact of RAG on the field of natural language processing.

2 RELATED WORK

This section provides essential background for understanding RAG (Retrieval-Augmented Generation). We begin by tracing the origin of RAG, exploring its evolution from its formal introduction and subsequent development. Moreover, we review existing surveys in the field, highlighting their focuses and limitations, which helps position our comprehensive survey in the current research landscape.

2.1 The Origin of RAG

As language models evolved, researchers began exploring ways to improve text generation through retrieval mechanisms [52, 244]. Nevertheless, during this time, scholars had not yet established a specific term for this methodology. By 2020, Facebook introduced the concept of RAG and effectively applied it to knowledge-intensive tasks [126]. RAG enhances generative models by integrating external knowledge through retrieval mechanisms, significantly improving performance in tasks such as question answering and text understanding. Simultaneously, Google’s REALM [81] adopted a comparable retrieval-enhanced approach. By incorporating a latent knowledge retriever during the pre-training phase, REALM achieved substantial advancements in open-domain question answering, establishing state-of-the-art results at the time.

Since then, RAG has gained growing research attention due to its unique advantages over traditional generative models. The retrieval-generation synergy enables models to dynamically access relevant information from external knowledge bases during complex reasoning tasks, thereby enhancing both the informativeness and accuracy of generated content. This paradigm represents a fundamental shift in natural language processing, where models no longer solely rely on their internal parameters but can effectively leverage vast external knowledge sources. As shown in Figure 1, while early works gradually explored different aspects of RAG since 2020, the field witnessed an exponential growth

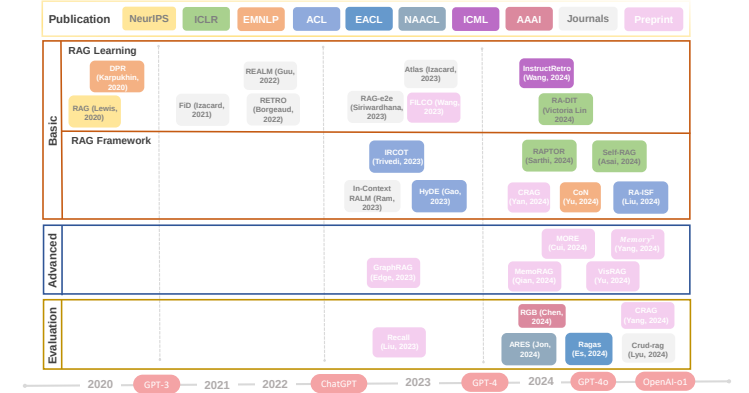


Fig. 1. A Framework for Organizing RAG Works. The timeline spans from 2020 to the present, categorizing RAG-related research into three main areas: Basic (including RAG Learning and RAG Framework), Advanced, and Evaluation. Key milestones in language models (GPT-3, ChatGPT, GPT-4) are marked along the timeline.

following the release of ChatGPT in late 2022, with numerous studies emerging to enhance large language models through retrieval-augmented approaches. These ongoing developments continue to shape the landscape of modern natural language processing.

2.2 Related Surveys

With the ongoing advancements in the field of generative AI, specifically in RAG, numerous surveys have emerged. However, these surveys often focus on specific aspects of the field. They either concentrate solely on a single foundation of RAG or provide a concise overview of enhancement methods for RAG in limited scenarios. Most existing works emphasize text-related RAG tasks supported by large language models without delving into other modalities. The survey by Li et al. [131] offers a fundamental overview of RAG and discusses specific applications within the realm of text generation tasks. Recent surveys by Gao et al. and Fan et al. [63, 73, 94] explore RAG in the context of large language models, with a specific emphasis on query-oriented RAG enhancement methods. Wu et al. [246] delves into key technologies of RAG in the areas of retrieval, while simultaneously introducing its broad applications in natural language processing tasks. Several other works investigate a more general approach to RAG. Recent Works by Zhao et al. [282, 283] extend RAG to multi-modal contexts, exploring its technologies and applications in broader AI-generated content (AIGC) scenarios. Another work by Peng et al. [184] examines how graph-structured information can aid more precise and comprehensive retrieval in RAG, enhancing relational knowledge acquisition and the generation of context-aware responses. In addition to concentrating on research pertaining to RAG technology, recent scholarship has increasingly directed its focus towards the assessment of RAG systems. A survey by Yu et al. [266] addresses the evaluation challenges

Table 1. Comparison of Different RAG Surveys. **LLM**: whether the survey discusses RAG in the context of large language models; **Multimodal**: whether it covers multimodal RAG; **Graph**: whether it discusses graph-structured information in RAG; **Advanced**: the coverage of advanced RAG techniques; **Evaluation**: whether it addresses evaluation methods; **Knowledge**: whether it takes a knowledge-centric perspective.

	Year	LLM	Multimodal	Graph	Advanced	Evaluation	Knowledge
Li et al. [131]	2022	✗	✗	✗	✗	✗	✗
Gao et al. [73]	2024	✓	✗	✗	Training	✗	✗
Fan et al. [63]	2024	✓	✗	✗	Training, Adaptive	✗	✗
Huang and Huang [94]	2024	✓	✗	✗	Training, Adaptive	✗	✗
Wu et al. [246]	2024	✓	✗	✗	Training, Adaptive	✗	✗
Zhao et al. [283]	2023	✓	✓	✗	✗	✗	✗
Zhao et al. [282]	2024	✓	✓	✗	✗	✗	✗
Peng et al. [184]	2024	✓	✗	✓	✗	✗	✗
Yu et al. [266]	2024	✓	✗	✗	✗	✓	✗
(Ours)	2025	✓	✓	✓	All	✓	✓

of RAG systems, providing an integrated evaluation framework and examining existing benchmarks along with their limitations. Despite covering multiple aspects of RAG, existing surveys still lack a comprehensive review that includes discussions on RAG foundations, enhancement methods, and their applications across different domains. Furthermore, current surveys generally overlook the essence of RAG—knowledge utilization. This paper aims to fill this gap through a systematic survey, exploring RAG with a knowledge-centric approach.

3 OVERVIEW OF RETRIEVAL-AUGMENTED GENERATION

Retrieval-Augmented Generation (RAG) has emerged as a powerful paradigm that combines the internal knowledge of Large Language Models (LLMs) with external knowledge to enhance generation quality. By integrating retrieval mechanisms into the generation pipeline, RAG addresses the limitations of traditional sequence-to-sequence frameworks, particularly in scenarios requiring extensive domain knowledge. This section presents a comprehensive overview of RAG, discussing its core components: **retrieval** from external knowledge sources, **generation** with both internal and external knowledge, and the critical process of **knowledge integration** that bridges these elements.

3.1 Problem Formulation

Most generation tasks can be conceptualized as a transformation from an input sequence x to an output sequence $y : y = f(x)$. However, this basic framework may be limited by insufficient information or complex contextual requirements. RAG addresses this limitation by introducing a retrieval component that enriches the input with relevant external knowledge. Specifically, a retrieval function g extracts relevant information z from external knowledge repositories: $z = g(x)$. The generation function f then combines the input sequence x with the retrieved context z to produce the output: $y = f(x, z) = f(x, g(x))$. This enhanced framework leverages external knowledge to improve generation quality, particularly beneficial for tasks requiring complex contextual understanding like machine translation and visual question answering.

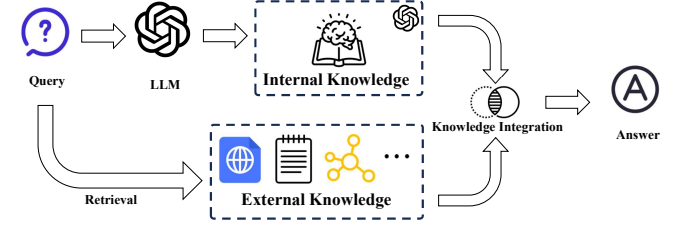


Fig. 2. Overview of RAG. The framework consists of three main components: (1) A query is processed by an LLM with its internal knowledge, (2) External knowledge is retrieved based on the query, and (3) Knowledge integration combines both internal and external knowledge to generate the final answer.

3.2 Retrieval

In context of Large Language Models (LLMs), the retrieval process in RAG aims to provide useful knowledge from external sources. Such knowledge may take multiple modalities, including text, image, video, audio, code, table, etc. Correspondingly, the storage formats are diverse, spanning from simple text files to complex databases and search engines. Additionally, the structure of this knowledge is heterogeneous; it may exist as unstructured plain text, as semi-structured html data or as structured graph data. Any process of obtaining relevant information from external knowledge bases can be considered as retrieval, irrespective of the modality, storage format, or structure of the knowledge.

3.3 Generation

In the generation process, generative models incorporate both internal and external knowledge to convert inputs into coherent and pertinent outputs. Regardless of the data format or task, the essence of the generation phase lies in "Denoising" [13, 241] and "Reasoning" [232, 291]. Denoising focuses on filtering out irrelevant or contradictory information from retrieved knowledge, ensuring that only reliable and relevant information influences the generation process. Meanwhile, reasoning enables the model to effectively synthesize information from multiple sources, draw logical connections, and produce well-founded outputs. Through effective "Denoising" of retrieved information and sophisticated "Reasoning" over multiple knowledge sources, generative models can handle complex tasks while maintaining output accuracy and coherence.

3.4 Knowledge Integration

Knowledge integration, often referred to as *augmentation* in the context of RAG, is a critical process that integrates internal knowledge from LLMs with retrieved external knowledge. This neural-symbolic integration can be achieved through three main approaches. Input-Layer Integration [97, 199] concatenates retrieved documents directly with the original query before feeding into the model, allowing simultaneous processing of both query and external knowledge. Output-Layer Integration [114] incorporates retrieved knowledge at the logits level to calibrate the model's final predictions, particularly effective for improving output accuracy. Intermediate-Layer Integration [23] integrates external

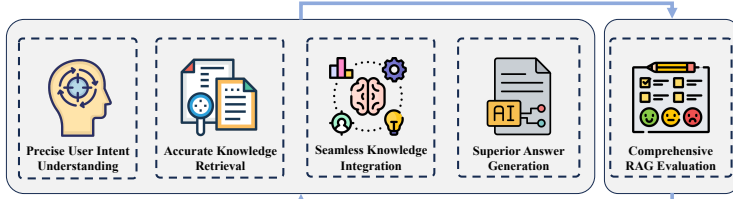


Fig. 3. Fundamentals and objectives of RAG, including user intent understanding, knowledge retrieval, knowledge integration, answer generation and RAG evaluation.

knowledge into the model’s hidden states during generation, enabling more nuanced interaction between internal and external knowledge representations. Each integration strategy offers distinct advantages and can be selected based on specific task requirements and computational constraints.

4 FUNDAMENTALS AND OBJECTIVES OF RAG

RAG models have emerged as a powerful paradigm that combines the strengths of information retrieval and answer generation. By leveraging external knowledge sources, RAG models enhance the quality and relevance of generated content, making them highly effective for various applications such as question answering, summarization, and conversational agents. However, the integration of retrieval and generation components introduces a complex set of characteristics and challenges that need to be addressed to fully realize the potential of RAG systems. This section examines the key aspects of RAG, exploring the main components of user intent understanding, knowledge retrieval, knowledge integration, answer generation, and evaluation metrics while providing deeper insights to each area, shown in Figure 3.

4.1 Precise User Intent Understanding

Accurately understanding user intent [28, 153, 275] is critical for RAG models to generate responses that are both semantically relevant and contextually appropriate, which requires analyzing user queries that may be ambiguous or contain implicit meanings, often lacking explicit details. User queries can vary widely due to user-specific personalized preferences posing challenges in determining the exact intent behind each request. Moreover, subtle language features—such as ambiguous expressions, informal phrases, or culturally specific references—can obscure the true intent of the user, making it harder for the LLM to extract the essential information. This complexity requires an approach to user intent understanding that goes beyond traditional keyword matching and instead delves into deeper semantic analysis to accurately capture user intent.

To overcome these challenges, RAG models need to implement advanced intent understanding strategies that incorporate contextual understanding and semantic analysis. Strategies such as text modeling [4], query rewrite [158], and intent clustering can enhance the LLM’s ability to interpret user needs, leading to more precise retrieval and generation. By continuously refining these techniques, RAG models can align with evolving user expectations and domain-specific requirements. Furthermore, precise intent understanding reduces the risk of irrelevant or vague

responses, thereby improving user satisfaction, which not only strengthens retrieval and generation processes but also lays a foundation for delivering adaptable, high-quality responses across diverse and complex real-world scenarios.

4.2 Accurate Knowledge Retrieval

The knowledge retrieval phase in RAG models is essential for providing relevant context for the generation process, thereby directly influencing the quality of the generated output. High retrieval accuracy ensures that the most relevant knowledge are accessed, while efficiency guarantees that the retrieval process does not become a bottleneck in real-time applications. One of the primary challenges in knowledge retrieval is handling large-scale datasets, which require sophisticated indexing techniques such as inverted indices, approximate nearest neighbor (ANN) [12] search, and scalable vector representations [77] to maintain swift access times. Additionally, the diversity and unstructured data sources demand robust preprocessing and normalization methods to ensure consistency and relevance.

Balancing precision and recall is another critical aspect. High precision minimizes the retrieval of irrelevant contents, thereby reducing noise in the generative process. Conversely, high recall ensures comprehensive coverage of relevant information, which is essential for tasks requiring in-depth understanding. Achieving the above balance often involves fine-tuning retrieval algorithms and leveraging relevance feedback strategies. Moreover, the dynamic nature of information sources, where data is frequently updated or expanded, poses ongoing challenges for maintaining up-to-date and accurate retrieval strategies. Addressing these issues requires continuous adaptation and optimization of retrieval strategies to handle evolving datasets accurately and effectively.

4.3 Seamless Knowledge Integration

Integrating retrieved knowledge into LLM is a challenging task that demands seamless alignment between external knowledge and the LLM’s internal knowledge [78]. Such integration is crucial for generating outputs that are not only semantically relevant but also coherent and contextually appropriate. One of the significant challenges is ensuring that the LLM can effectively interpret and utilize the retrieved information without introducing inconsistencies or factual inaccuracies, which requires strategies for merging diverse data types, such as text chunks, and structured data into a unified representation that LLM can process.

Furthermore, the integration process needs to handle potential conflicts between retrieved knowledge and the LLM’s pre-existing knowledge base. For instance, discrepancies or contradictions between newly retrieved information and the LLM’s training data can lead to confusion and reduced output quality, which requires advanced techniques for conflict resolution and knowledge validation to ensure that the integrated information enhances rather than detracts from the generation process. Additionally, the temporal relevance of retrieved knowledge needs to be considered, particularly for applications requiring up-to-date information. This necessitates flexible integration strategies that prioritize recent and contextually relevant data, thereby maintaining the relevance and accuracy of the generated content over time.

4.4 Superior Answer Generation

Achieving superior answer generation in RAG models involves the complex task of synthesizing retrieved information with the LLM’s natural language abilities to ensure outputs are accurate, relevant, and coherent. Such integration requires a refined alignment between external knowledge sources and the LLM’s internal knowledge, reducing potential discrepancies that could lead to factual inconsistencies. Cross-referencing and validation strategies [241] play a vital role here, acting as safeguards that help verify the accuracy of information and prevent the spread of misinformation.

With these checks in place, the LLM can more reliably produce content that aligns with verified sources, enhancing the user's trust in the generated output.

Another core challenge lies in maintaining fluency and naturalness in the generated text, especially when merging information from diverse sources that may vary significantly in style or structure. The LLM must blend these inputs seamlessly to create outputs that are not only readable but contextually appropriate for the user. Advanced language modeling techniques and a strong understanding of context are crucial to achieving coherence, allowing the LLM to navigate and integrate complex or varied information effectively. Additionally, the generation component must remain adaptable, capable of adjusting dynamically to different contexts and user intents. Whether delivering precise answers, summarizing extensive data, or engaging in interactive dialogue, the LLM should adapt tone, style, and content to suit the situation. This versatility demands robust training strategies that enable the model to handle diverse scenarios.

4.5 Comprehensive RAG Evaluation

Evaluating the performance of RAG presents a unique set of challenges due to the dual nature of retrieval and generation tasks. Traditional evaluation metrics, such as BLEU [181] and ROUGE [138], primarily focus on the quality of the generated text by comparing it to reference outputs. However, these metrics may not adequately capture the effectiveness of the retrieval component, which plays a crucial role in determining the relevance and accuracy of the generated content. To address this, comprehensive evaluation frameworks must integrate metrics that assess both retrieval accuracy (e.g., precision, recall, F1-score [256]) and generation quality (e.g., coherence, fluency, factual accuracy).

Developing such comprehensive evaluation metrics is inherently challenging, as it requires balancing quantitative measures with qualitative assessments to provide a well-rounded evaluation of the system's performance. Additionally, aligning these metrics with real-world user satisfaction and practical utility adds another layer of complexity. Subjective factors, such as the perceived relevance and helpfulness of the generated responses, are difficult to quantify but are essential for evaluating the true effectiveness of RAG systems. Incorporating user studies, human evaluations, and adaptive metrics that evolve with user feedback can enhance the accuracy and relevance of performance assessments. Furthermore, ensuring that evaluation metrics remain robust across different domains and applications is critical for maintaining the generalizability and applicability of RAG systems in diverse real-world scenarios. Additionally, the citation generation in RAG is crucial for verifying the credibility and traceability of the retrieved information in the evaluation process.

Conclusion. RAG systems include many different characteristics and challenges that span the entire process of intent understanding, knowledge retrieval, knowledge integration, answer generation, and evaluation. Each component presents its own set of complexities, from handling large-scale and dynamic datasets in retrieval to ensuring seamless integration and high-quality generation of content. Addressing these challenges requires innovative solutions that leverage advanced algorithms, robust training methodologies, and comprehensive evaluation frameworks. Moreover, the interdependent nature of retrieval and generation tasks in RAG systems necessitates a complete approach to system design and optimization, ensuring that improvements in one area work together to improve overall performance. Continued research and development in these areas hold significant promise for advancing the capabilities of RAG models, enabling more intelligent, accurate, and contextually aware generative systems that can meet the evolving demands of diverse applications and industries.

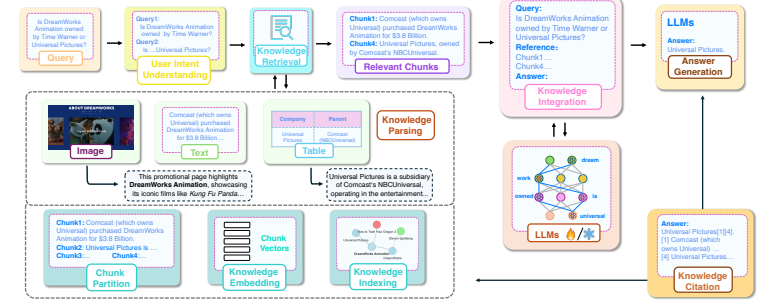


Fig. 4. Basic RAG Approaches, including multi-source knowledge, embedding, indexing, retrieval and generation.

5 BASIC RAG APPROACHES

RAG models leverage external knowledge to enhance the generation process, enabling more accurate and contextual responses. The basic RAG approaches consists of several key steps: user intent understanding, knowledge source and parsing, knowledge embedding, knowledge indexing, knowledge retrieval, knowledge integration, answer generation, and knowledge citation. This section explains these core components, showing how RAG systems understand user queries, process different types of knowledge, convert information into vectors, build search indexes, retrieve relevant content, integrate knowledge with models, and generate answers with proper citations.

5.1 User Intent Understanding

High-quality queries are essential for retrieving valuable knowledge. Since users' intentions are often ambiguous, accurately interpreting user queries is crucial for enabling more effective and precise retrieval [34, 35]. Currently, many studies focus on enhancing the understanding of user queries. This subsection primarily discusses two key methods for improving query quality: query decomposition and query rewriting.

5.1.1 Query Decomposition. Query decomposition methods have emerged as effective strategies to enhance the reasoning abilities of language models, particularly for complex tasks requiring multi-step or compositional reasoning. Techniques such as least-to-most prompting[287] break down difficult problems into progressively simpler subproblems, improving generalization to harder tasks. This approach has demonstrated remarkable performance on tasks like SCAN, where models like GPT-3 achieved over 99% accuracy with just 14 exemplars. Self-ask[190] takes a similar approach but further refines the process by having the model ask and answer follow-up questions, reducing the compositionality gap and enabling better multi-hop reasoning. Additionally, chain-of-verification (CoVe)[51] improves the reliability of answers by having the model verify its responses through independent fact-checking, significantly reducing hallucinations in tasks like list-based questions and long-form generation. Another promising method is search-in-the-chain (SearChain)[254], which integrates information retrieval (IR) into the reasoning process. In this framework, the model constructs a Chain-of-Query (CoQ), where each query is verified using IR, enhancing both the

accuracy and traceability of the reasoning path. SearChain allows the model to dynamically adjust its reasoning based on retrieved information, improving performance on knowledge-intensive tasks such as multi-hop QA and fact-checking.

5.1.2 Query Rewriting. Query rewriting has become a crucial technique for enhancing the performance of RAG, particularly in addressing semantic gaps and improving task outcomes. Rewrite-Retrieve-Read (RRR)[158] optimizes the query generation process by using an LLM to create and refine the query before retrieval, with a trainable rewriter guided by reinforcement learning. This approach improves the alignment between queries and target knowledge, boosting performance in open-domain QA and multiple-choice tasks. BEQUE[186] focuses on long-tail queries in e-commerce search, bridging semantic gaps through supervised fine-tuning, offline feedback, and contrastive learning, resulting in significant improvements in business metrics like GMV and transactions. HyDE[71] introduces a zero-shot approach to dense retrieval, where an instruction-following LLM generates a hypothetical document, which is then encoded and used to retrieve relevant documents, outperforming traditional unsupervised retrievers. Step-Back Prompting [285] encourages LLMs to abstract high-level concepts from specific examples, improving reasoning on tasks like STEM, multi-hop QA, and knowledge-based reasoning. These methods collectively enhance the effectiveness and scalability of RAG in knowledge-intensive tasks across diverse domains.

5.2 Knowledge Source and Parsing

The types of knowledge that RAG can utilize are diverse, providing extensive contextual information for LLMs. However, the parsing and efficient integration of these heterogeneous knowledge sources pose significant challenges. This subsection introduces the categories of knowledge employed by RAG, including structured, semi-structured, unstructured, and multimodal knowledge, along with their respective parsing and integration methodologies.

5.2.1 Utilization of Structured Knowledge. Knowledge graphs (KGs) are structured representations that encapsulate entities and their interrelations in a graph format, offering advantages for RAG. Their structured nature facilitates efficient querying and retrieval, while the semantic relationships they capture enable nuanced understanding and reasoning[93, 103]. Additionally, KGs integrate information from diverse sources, providing a unified knowledge base. However, integrating KGs into RAG systems presents challenges, including the complexity of navigating and extracting relevant subgraphs from extensive KGs, scalability issues as KGs expand, and aligning structured data with the unstructured data processing of language models.

Recent research has addressed these challenges through innovative approaches. For instance, GRAG[92] retrieves textual subgraphs across multiple documents, enhancing information retrieval efficiency in RAG systems. KG-RAG[211] incorporates the Chain of Explorations (CoE) algorithm to improve knowledge graph question answering (KGQA) tasks by efficiently navigating KGs. GNN-RAG[163] utilizes Graph Neural Networks (GNNs) to retrieve and process information from KGs, enhancing reasoning capabilities before interfacing with LLMs. Constructing KGs from historical data serves as an external knowledge source for RAG, effectively enhancing information retrieval and generation[255]. The SURGE framework[110] leverages KG information to generate contextually relevant and knowledge-grounded dialogues, improving interaction quality. Domain-specific implementations, such as SMART-SLIC[17], KARE[104], ToG 2.0[157], KAG[137], have demonstrated the efficacy of KGs as external knowledge sources in specialized domains, enhancing accuracy and efficiency in RAG systems. These advancements underscore the potential of KGs to enhance global retrieval capabilities and the summarization of extensive documents, thereby improving both the accuracy and efficiency of RAG systems. Notably, KGs play a specialized role in RAG systems within certain specific domains, offering tailored solutions to domain-specific challenges[221].



Fig. 5. Diverse knowledge utilized by RAG, including structured, semi-structured, unstructured and multimodal knowledge.

Tables convey rich information through the interplay of rows and columns, enhancing data density but posing challenges for language models to interpret[239]. Tables often encapsulate large volumes of data succinctly, playing a crucial role in summarizing facts and quantitative data[284]. Additionally, the inherent organization of tables supports straightforward integration with various data processing tools and algorithms. However, integrating tabular data into RAG systems presents several challenges. One significant issue is the semantic understanding of table content, as tables may contain implicit relationships and context that are not immediately apparent. Another challenge is the diversity in table structures across different domains, which complicates the development of generalized models for table interpretation[222]. Furthermore, translating natural language queries into structured queries that can effectively interact with databases requires sophisticated natural language processing techniques.

Recent research has tackled these challenges through the development of innovative methodologies. For instance, TableRAG[29, 48] utilizes query expansion along with schema and cell retrieval to accurately identify crucial information before passing it to language models. This approach allows for the efficient handling of large-scale tabular data, improving the accuracy and relevance of generated responses. Additionally, models like TAG[29, 48] and extreme-RAG[206] unify Text-to-SQL[257, 269, 273] and RAG methods, broadening the range of user queries that can be addressed. These advancements underscore the potential of leveraging tabular data in RAG systems to enhance information retrieval and generation capabilities. By effectively integrating structured data, these systems can provide more precise and contextually relevant responses, thereby improving user experience and satisfaction.

5.2.2 Extraction of Semi-Structured Knowledge. Semi-structured data occupies a middle ground between structured and unstructured formats, featuring organizational elements without a rigid schema. Examples include JSON and XML files, emails, and notably, HTML documents. HTML serves as the foundation of web pages, blending structured components like tags and attributes with unstructured content such as free-form text. This hybrid nature allows for the representation of complex information, including text, images, and links, in a hierarchical format. However, the flexibility of HTML can lead to inconsistencies and irregularities, posing challenges for data extraction and integration into RAG. Currently, research in RAG has started to focus on parsing HTML, as seen in approaches like HtmlRAG[226]. Nevertheless, in the majority of scenarios, open-source HTML parsing technologies remain indispensable for efficient data extraction and seamless integration. These tools provide robust parsing capabilities and adaptability to diverse HTML structures, ensuring high efficiency and accuracy in various application contexts.

To effectively parse HTML, various techniques and open-source libraries have been developed, aiming to convert HTML content into structured formats like Document Object Model (DOM) trees, enabling efficient data manipulation and retrieval. Notable libraries include Beautiful Soup[203], a Python library that parses HTML and XML documents, creating parse trees for easy data extraction; html5ever[215], an open-source HTML parser developed by the Servo

project, adhering to the WHATWG "HTML5" specification; `htmlparser2`[21], a robust HTML parser for Node.js environments, offering a fast and flexible way to process HTML documents; `MyHTML`[24], a high-performance HTML5 parser based on the Crystal language, binding to the `lexborisov`'s `myhtml` and `Modest` libraries; and `Fast HTML Parser`[39], an extremely fast HTML parser that generates a minimal DOM tree and supports basic element querying. These tools exemplify advancements in parsing semi-structured data, particularly HTML, facilitating more efficient data extraction and integration into various applications, including RAG systems.

5.2.3 Parsing of Unstructured Knowledge. Unstructured knowledge encompasses a broad array of data types that lack consistent structure, such as free-form text, and PDF documents. Unlike structured data, which adheres to predefined schemas, unstructured data varies widely in format and often contains complex content, making direct retrieval and interpretation challenging. Among these unstructured formats, PDF documents are particularly prevalent in knowledge-intensive domains, including academic research, legal documents, and corporate reports. PDFs often house a wealth of information, with elements like text, tables, and embedded images, but their inherent structural variability complicates extraction and integration into RAG systems.

Parsing PDFs remains challenging due to the need to accurately interpret diverse layouts, fonts, and embedded structures. Converting PDFs into machine-readable formats for RAG systems requires OCR to capture text, layout analysis to understand spatial relationships, and advanced methods to interpret complex elements like tables and formulas. Modern OCR technologies have improved this process by integrating visual and linguistic cues, as seen in vision-language models like `Levenshtein OCR`[44] and `GTR`[87]. Frameworks like `OmniParser`[236] and `Doc-GCN`[152] offer unified approaches for preserving structure, and language models such as `ABINet`[64] enhance OCR with bidirectional processing for greater accuracy. These advancements illustrate a shift toward more sophisticated, multi-modal OCR solutions for interpreting unstructured data in RAG systems. In parallel, the development of open-source tools for PDF-to-Markdown conversion has provided additional support for unstructured knowledge integration. These tools cater to various document structures and use cases, offering specialized capabilities for handling complex PDF layouts. For example, `GPTPDF`[41] employs visual models to parse intricate elements, such as tables and formulas, into structured Markdown, which is highly cost-efficient for large-scale processing. `Marker`[183] focuses on cleaning noisy elements while preserving the document's core formatting, making it ideal for academic and scientific documents. `PDF-Extract-Kit`[180] supports high-quality content extraction, including formula recognition and layout detection, while `Zerex OCR`[74] converts PDF pages into images and uses GPT models to generate Markdown, effectively managing even complex document structures. `MinerU`[179] is a comprehensive solution that retains the original document structure, including titles and tables, and supports OCR for corrupted PDFs. `MarkItDown`[165] is a versatile tool that converts a wide range of file types, including PDFs, media, web data, and archives, into Markdown. These tools reflect the trend towards more accurate, structured solutions for extracting valuable information from unstructured PDF data.

5.2.4 Integration of Multimodal Knowledge. Multimodal knowledge—including images, audio, and video—offers rich, complementary information that can significantly enhance RAG systems, especially for tasks requiring deep contextual understanding. Images provide spatial and visual details; audio contributes temporal and phonetic layers; and video combines both spatial and temporal dimensions, capturing motion and complex scenes. Traditional RAG systems, primarily designed for text-based data, often struggle to process and retrieve information from these modalities, leading to incomplete or less nuanced responses when non-textual content is essential.

To address these limitations, modern multimodal RAG systems have developed basic approaches to integrate and retrieve data across various modalities. The fundamental idea is to align different modalities into a shared embedding

Knowledge Form	Embedding Models	Dimensions
Text	Sparse Encoders	Bag of Words (BoW) N-gram TF-IDF Size of the Vocabulary
	Traditional Dense Encoders	Word2vec[166, 167] GloVe[187] fastText[22] Typically 100, 200, or 300 50, 100, 200, or 300 300 (Default)
	BERT Architecture Pre-training Encoders	BERT[50] RoBERTa[148] ALBERT[121] DistilBERT[210] ELECTRA[38] DPR[111] 768 (Base) 768 (Base) 768 (Base) 768 768 (Base) 768
	LLM-based Encoders	BGE[130] NV-Embed[124] SFR-Embedding[207] text-embedding[178] 4096 4096 4096 768
	Specific Domain Encoders	ru-en-RoSBERTa[219] USTORY[263] E5-Base-4k[289] 1024 768 768
Image	Vision Transformer (ViT)[55] CLIP[196]	768 (Base) 512
Audio	Wav2Vec 2.0[16] CLAP[247]	768 (Base) 512
Video	ViViT[9]	768 (Base)

Table 2. Knowledge Embedding Models

space for unified processing and retrieval. For example, models like `CLIP`[195] align vision and language in a shared space, while models for audio like `Wav2Vec 2.0`[16] and `CLAP`[247] focus on audio-text alignment. For video, basic approaches like `ViViT`[9] handle both spatial and temporal features. These foundational techniques provide the basis for more advanced multimodal RAG applications, which will be discussed in detail in the advanced approaches section.

5.3 Knowledge Embedding

Knowledge is conventionally stored in extensive textual documents, which are first segmented into concise and meaningful units, each encapsulating a distinct core idea. These units are subsequently transformed into vector embeddings that encode semantic information, facilitating efficient retrieval through similarity metrics. In this subsection, we will introduce chunk partitioning for segmenting knowledge into meaningful units and delves into multimodal embedding models, including text, images, audio, and video. Table 2 provides a summary of the corresponding models utilized in RAG.

5.3.1 Chunk Partitioning. Chunk partitioning is a foundational process that significantly impacts document retrieval quality and, consequently, the overall performance[218, 270]. The primary objective of chunk partitioning is to segment large bodies of text into manageable units, or "chunks," which enables more efficient retrieval and generation[235]. By dividing complex documents into smaller, coherent units, RAG can achieve finer-grained control over context

preservation and enhance information retrieval accuracy. The goal and challenges of chunk partitioning in RAG involve ensuring that segmented chunks retain meaningful context while avoiding redundancy and information loss. Traditional chunking methods, such as fixed-length, rule-based or semantic-based partitioning, are relatively straightforward but lack the flexibility to capture nuanced structures in complex texts. These methods often struggle with documents that have varied formats or contain cross-paragraph semantic flows, resulting in suboptimal retrieval due to fragmented contextual information[122, 128, 149, 155, 249, 277].

Evolving chunk partitioning methods have shifted from early fixed-length approaches toward more adaptive and intelligent strategies. Recent optimization strategies aim to capture fine-grained information distribution and semantic structure within text. For example, chunking at the "proposition level" breaks text into small units of single facts, allowing for richer information capture[31]. Other methods employ LLMs for dynamic chunk partitioning, such as the LumberChunker approach[56], which uses LLMs to detect shifts in content across paragraphs, thereby creating context-sensitive chunks. Additionally, meta-chunking methods[280] like Margin Sampling Chunking and Perplexity Chunking refine chunk boundaries to better suit RAG requirements, improving context capture and retrieval effectiveness. An innovative approach, late chunking[83], deviates from conventional practices by embedding entire documents before partitioning them. This enables the model to retain complete context, leading to improved retrieval outcomes, especially for complex or contextually dense texts. These advanced chunking strategies empower RAG to better adapt to complex document structures, capture nuanced information, and enhance retrieval accuracy, paving the way for more robust support in diverse retrieval and generation tasks.

5.3.2 Text Embedding Models. When retrieving chunks based on a query, the process relies on the computation of vector similarity (e.g., cosine similarity) between the query and the chunks. Accurate mapping of chunks to vector representations is critical for effectively capturing semantic meaning and aligning with relevant queries. Traditional methods, such as Bag of Words (BoW), emphasize word frequency but disregard syntax, while N-gram captures linguistic structure but suffers from dimensionality challenges. The TF-IDF model incorporates both word frequency and context yet remains constrained by high-dimensionality issues. These approaches fail to comprehensively represent semantic information. In contrast, modern word embeddings, typically based on deep learning, employ methods such as word2vec[166, 167], GloVe[187], and fastText[22], which encode words by leveraging context, co-occurrence patterns, or root variations. However, these embeddings are static and unable to adapt to varying contexts. The introduction of the transformer architecture[234], particularly BERT[50], represented a significant advancement in capturing contextual meanings and enhancing semantic representation. BERT[50] and its derivatives, including RoBERTa[148], ALBERT[121], and DPR[111], have substantially improved document retrieval, with DPR[111] being employed in the original RAG[127] for retrieving relevant documents.

LLMs now dominate embedding methodologies, characterized by billions of parameters trained on extensive corpora, offering advanced semantic understanding and contextual reasoning. Prominent models include BGE[130], NV-Embed[124], and SFR-Embedding[207], which demonstrate strong performance across multilingual and specialized benchmarks. BGE[130] excels in multilingual benchmarks, effectively encoding semantically rich queries. Similarly, NV-Embed[124] has gained recognition for achieving record-breaking results in the Massive Text Embedding Benchmark (MTEB)[170], securing top rankings across various tasks. Beyond general-purpose embeddings, domain-specific models are increasingly fine-tuned to address specialized tasks. For instance, ru-en-RoSBERTa[219] is tailored for Russian-language tasks, WE-iMKVec[116] focuses on healthcare-related text embeddings, USTORY[263] identifies

relevant news topics, and E5-Base-4k[289] explores encoding strategies for long-context retrieval. The fine-tuning of these models for specific domains enables RAG systems to achieve superior performance in targeted applications.

A wide variety of embedding models are available for selection. Consequently, the challenge lies in identifying the most suitable model and determining an appropriate method for storing embedding vectors. To facilitate model selection, benchmarks such as MTEB[170], C-MTEB[250], LongEmbed[289], SEB[60], and AIRBench[14] provide comprehensive evaluations across diverse tasks and domains. MTEB[170], for example, includes 113 languages and 457 models, offering a broad comparison of embedding performance across tasks such as retrieval and re-ranking. C-MTEB[250] focuses on Chinese-language embedding capabilities, LongEmbed[289] evaluates performance in long-context retrieval, and SEB[60] assesses multilingual text embeddings. AIRBench[14] is specifically designed for RAG, evaluating retrieval and re-ranking performance across domains such as law, healthcare, and news. In practical applications, vector databases, including FAISS, Milvus, and Pinecone, are commonly employed to store and retrieve high-dimensional vectors. These databases facilitate fast and efficient similarity search based on embedding vectors, and when integrated with RAG models, they significantly improve retrieval efficiency and accuracy, particularly in large-scale corpora.

5.3.3 Multimodal Embedding Models. Knowledge is represented not only in text but also in images, audio, and video. Consequently, there is an increasing demand for multimodal embedding models that integrate information from various modalities into a cohesive vector space. These models are specifically designed to capture the relationships and shared information across different data types, thereby enabling more comprehensive and unified representations.

For images, models process image formats such as JPG or PNG to generate embeddings within the same semantic vector space as text. Normalizer-Free ResNets (NFNet)[25] provides an efficient framework for extracting image features, while the Vision Transformer (ViT)[55] leverages the transformer architecture to learn high-quality representations. Contrastive Language–Image Pretraining (CLIP)[196] further advances the field by aligning visual and textual modalities through contrastive learning, producing versatile embeddings for applications like zero-shot classification and cross-modal retrieval. CLIP[196] has become a cornerstone of modern multimodal AI systems.

For audio, models extract essential features such as pitch, timbre, rhythm, and semantics, enabling effective and meaningful audio analysis for retrieval tasks. Wav2Vec 2.0[16], a self-supervised learning model, learns audio representations directly from raw waveforms, producing rich, high-level embeddings suitable for diverse audio tasks. Inspired by CLIP[196], Contrastive Language–Audio Pretraining (CLAP)[247] is a SOTA model that generates audio embeddings by learning from paired audio and text data, offering a unified framework for integrating audio with natural language.

For video, models aim to represent video data in compact, feature-rich vectors that capture spatial, temporal, and semantic information. Video Vision Transformer (ViViT)[9], based on ViT[25], can effectively handle video understanding tasks by capturing both spatial and temporal features. Nowadays, VideoPrism[281] draws attention since it achieves SOTA performance across a wide range of video understanding benchmarks. It is particularly notable for its ability to generalize well across different video domains without needing task-specific fine-tuning.

In summary, good embeddings are essential to the search quality and search efficiency of retriever. Nowadays some text embedding models such as BGE[130] and NV-Embed[124] perform very well, which can dominates across almost all tasks. However, with the rapid development of multi-modal RAG, there is still an increasing requirement for better multimodal embedding models.

5.4 Knowledge Indexing

In RAG, an index is defined as a structured organization of data, which enables efficient access and retrieval of information from large-scale datasets. Index maps user queries to relevant document chunks, knowledge snippets, or other informational content, functioning as a bridge between the stored data and the retrieval mechanism. The effectiveness of indexing is crucial for RAG systems, as it directly impacts response accuracy, retrieval speed, and computational efficiency.

5.4.1 Index Structure. Structured Index. Structured indexing organizes data based on predefined, fixed attributes, typically following a tabular or relational format. In early knowledge retrieval efforts, such as in systems like **REALM**, textual inverted indexing was widely employed as a fundamental technique[82], while Table RAG utilizes a table-specific indexing structure that combines column-based and row-based indices to efficiently retrieve relevant table entries for language generation tasks[30].

Unstructured Index. Unstructured indexing, in contrast, is designed for free-form or semi-structured data, which is more commonly used in modern RAG systems. Vector indexing uses vectors obtained from previous embedding stages to improve retrieval efficiency, such as naive RAG[126], ANCE[252] and G-retriever which uses language models to convert textual attributes of graphs into vectors[86].

Graph index, a form of unstructured indexing, leverages the inherent strengths of graph structures to represent and retrieve interconnected data. In graph indexing, data points are depicted as nodes, while the relationships between these points are illustrated through edges. This indexing paradigm is particularly adept at capturing semantic relationships and contextual information, which are crucial for facilitating effective retrieval in RAG systems. Due to these advantages, graph indexing is increasingly being adopted in RAG applications. For instance, **ToG** directly utilizes the Knowledge Graph as its indexing structure[225], and further adds unstructured text data in **ToG 2.0**[156]. **Graph RAG** implies LLM to extract entities and relationships as the textual attributes of the graph, then divides its graph into sub-communities, creating text summaries for each community[57]. **RAPTOR** constructs a tree with different levels of summarization to enhance the overall understanding of long texts through recursive embedding, clustering, and summarizing text chunks[212]. While **GRAFT-Net** and **PullNet** fusing textual data from the corpus with the knowledge graph to construct heterogeneous graphs, and subsequently extracting sub-graphs to derive answers[223, 224]. Once the graph is established, a lot of graph search algorithms, such as BFS and HNSW[161], can be applied for subsequent Retrieval step.

Hybrid Index. In practice, a hybrid indexing approach typically performing better than a single one, for it combines structured and unstructured techniques to optimize retrieval across varied data types[185]. **ColBERT** uses an inverted index structure to quickly filter relevant documents, after which the similarity between the query and the candidate document embedding is computed via a vector index structure, which ensures high semantic accuracy while realizing fast retrieval[115].

5.4.2 Index Updating & Storage. Although less discussed, index updating plays a critical role in ensuring that RAG systems remain accurate and efficient when new data is introduced. In practice, index updates are typically performed through two primary approaches: **incremental updates** and **periodic reconstruction**, among which incremental updates, the process of adding or modifying entries within an existing index without the necessity of rebuilding the entire structure, allows for efficient adaptation to changing data while minimizing computational overhead, and thus is particularly beneficial in dynamic systems where new information is continuously introduced, such as **Light RAG**[79] and **REALM**[82].

Moreover, it is the storage of indexes that also plays a fundamental role in RAG models, which is usually influenced by the structure of indexes and will, directly or indirectly, influence the performance of the retrieval step. For vector indexes, **Chroma** utilizes a distributed architecture and is horizontally scalable for mass storage. **Faiss** has excellent similarity search capabilities, providing an optimized indexing method for efficient retrieval of nearest neighbors[107] and **Weaviate** uses graph structures for its data storage and integrates machine learning models to optimize vector storage and search[240]. For graph indexes, there are many well-established graph databases available, such as **JanusGraph** and **HugeGraph**, which use Gremlin as their query language[7, 98], as well as **Neo4j**, the largest and most recognizable one which uses Cypher[175].

5.5 Knowledge Retrieval

Knowledge Retrieval is the process of identifying and retrieving relevant knowledge from a vector database based on a given query. This process includes both retrieval strategies, which focus on how to identify relevant knowledge, and search methods, which define the algorithms used to quickly locate the most relevant information.

	Type	Models	Deep Learning
Retrieval Strategies	Sparse Retrieval	BM25[204]	×
		TF-IDF[205]	×
		likelihood [120]	×
		uniCOIL [139]	✓
		TILDE[290]	✓
		SPLADE[68]	✓
	Dense Retrieval	DPR [112]	✓
		ANCE [252]	✓
		Llama2vec [129]	✓
		RepLLaMA [159]	✓
	Hybrid Retrieval	RAP-Gen[238]	✓
		BlendedRAG[213]	✓
		ReACC[151]	✓
		BASHEXPLAINER[265]	✓
Search Approaches	Nearest Neighbor Search	Brute Force Algorithm	×
		KD Tree [19]	×
		Ball Tree [54]	×
		M-tree[37]	×
		Locality-Sensitive Hashing[6, 45]	×
	Approximate Nearest Neighbor Search	Spectral Hashing[242]	×
		Deep Hashing[143]	✓
		K-means tree[228]	×
		ANNOY	-
		HNSW [161]	×
		Product Quantization[99]	×

Table 3. Knowledge Retrieval Models

5.5.1 Retrieval Strategies. The objective of retrieval is to identify and extract the most relevant knowledge based on the input query. Specifically, the task involves retrieving the top-k most relevant chunks by employing a similarity function. Based on different similarity functions, retrieval strategies can be categorized into three types: sparse retrieval,

dense retrieval, and hybrid retrieval. *Sparse Retrieval.* Sparse retrieval strategies leverage sparse vectors to retrieve documents or chunks through term analysis and matching. Traditional sparse retrieval strategies utilize term matching metrics such as BM25[204], TF-IDF[205] and query likelihood[120], which estimate the relevance of documents to queries by calculating the frequency of term occurrences and inverse document frequency. Despite simplicity, these strategies remain prevalent in RAG due to their computational efficiency. Recently, some research explored the use of transformer language models as sparse retrievers. Models such as uniCOIL[139], TILDE[290] and SPLADE[68] encode text into high-dimensional sparse vectors, effectively representing documents in a bag-of-words fashion.

Dense Retrieval. Dense retrieval strategies encode both queries and documents into a low-dimensional vector space, where relevance is measured by the dot product or cosine similarity between their vector representations. Dense retrievers, such as DPR [112] and ANCE [252], are based on BERT architecture pre-training language model and fine-tuned on unsupervised data to generate high-quality representations for queries and documents. More recently, LLMs have shown remarkable performance in terms of semantic understanding and representational power. Based on the powerful capabilities of LLMs, efforts have been made to generate discriminative text embeddings using LLMs[129, 159, 169, 174, 278]. Llama2Vec[129] is a lightweight method that adapts LLMs for dense retrieval using two unsupervised pretraining tasks, EBAE and EBAR. RepLLaMA[159] fine-tunes LLaMA as a dense retriever, leveraging its capability to represent long documents holistically for efficient text retrieval.

Hybrid Retrieval. Hybrid retrieval strategies combine both sparse and dense retrieval techniques, aiming to optimize performance by leveraging the strengths of each approach. RAP-Gen[238] and BlendedRAG[213] integrate traditional keyword matching with deep semantic understanding, allowing systems to benefit from the efficiency of sparse retrieval while capturing deeper context through dense representations. BASHEXPLAINER[265] employs a two-stage training strategy that first utilizes a dense retriever to capture semantic information, followed by a sparse retriever to acquire lexical information, resulting in automatic code comment generation that performs well. This dual strategy addresses the limitations of each individual method; for instance, sparse strategies may struggle with semantic nuances, while dense strategies can be computationally intensive. By leveraging the strengths of both, hybrid models enhance retrieval accuracy and relevance across various tasks.

5.5.2 Search Approaches. Search approaches refer to algorithms designed to efficiently identify similar vectors from the vector database for a given query vector. Search approaches can be divided into two types: Nearest Neighbor Search (NNS) and Approximate Nearest Neighbor Search (ANNS). NNS aims to find the vector in a given set that is closest (or most similar) to the query vector, while ANNS is a variation of NNS that allows for a controlled degree of error or approximation in the search results[84].

Nearest Neighbor Search. The brute force algorithm for NNS is a straightforward algorithm that exhaustively scans all vectors in the database, calculating the distance to the query vector to identify the closest one. However, this method is computationally expensive and impractical for large-scale datasets. To mitigate this limitation, tree-based methods have been introduced to enhance search efficiency. For example, Bentley[19] proposed a method based on the k-d tree, which recursively partitions k-dimensional space into hyperrectangular regions, improving both data organization and search speed. Other tree-based structures, such as Ball-tree[54, 145, 177], R-tree[80], and M-tree[37] have also been used to enhance nearest neighbor search by partitioning the data into various structures, such as hyperspheres, rectangles, or metric spaces, thereby improving search performance, particularly in high-dimensional and complex datasets.

Approximate Nearest Neighbor Search. ANNS strike a balance between accuracy, speed, and memory efficiency, making it particularly useful for large-scale and high-dimensional data[84]. This involves hash-based methods, tree-based methods, graph-based methods, and quantization-based methods. Hash-based methods, such as Locality-Sensitive Hashing[6, 45], Spectral Hashing[242], and Deep Hashing[143], convert high-dimensional vectors into binary codes, optimizing memory usage and accelerating search operations. For instance, Deep Hashing[143] employs deep neural networks to learn hash functions that map high-dimensional vectors into binary codes while preserving the semantic relationships between similar data. Tree-based ANNS methods, including K-means tree[228] and ANNOY, organize data hierarchically, reducing the search space by efficiently traversing tree structures. These methods divide the dataset into partitions or clusters, such that only relevant regions are explored during the search. Graph-based methods, such as Hierarchical Navigable Small World (HNSW) [161], connect data points by edges that reflect their proximity to one another, allowing for fast nearest neighbor searches by navigating through the graph. Finally, quantization-based methods, such as Product Quantization[99], aim to compress the data by quantizing the vectors into a smaller codebook, thus reducing memory requirements while maintaining a good balance between search speed and accuracy. Overall, these diverse ANNS approaches provide powerful solutions for fast and efficient nearest neighbor search in large-scale, high-dimensional datasets, each with its own trade-offs in terms of accuracy, speed, and memory usage.

5.6 Knowledge Integration

Knowledge Integration refers to integrate the retrieved external knowledge with the internal knowledge of the generation model to enhance the output accuracy and coherence. Basically, there are three types of knowledge integration: input layer integration, intermediate layer integration, and output layer integration.

5.6.1 Input Layer Integration. Input Layer Integration refers to the direct integration of retrieved information with the original query at the input layer, aiming to enhance the contextual information during the generation process. Based on the method of integration, input layer integration can be categorized into two types: text-level integration and feature-level integration.

A simple method of text-level integration involves directly concatenating the retrieved top-k documents with the query. To reduce the impact of low-quality information and better leverage the in-context learning capability of LLMs, some approaches[76, 133] rerank the chunks to prioritize the most relevant content, while others[8, 154] apply weighted filtering to remove irrelevant information from the retrieved content. Due to the input length limitations of LLMs, some methods [102, 144, 253] compress the context to enable the model to learn more information within the available input size. These approaches aim to retain the most relevant content while reducing the overall length, ensuring that critical information is preserved for effective generation.

Feature-level integration focuses on integrating the encoded forms of retrieved content with the original input at the feature level. Rather than simply concatenating raw texts, this method[46, 96] transforms both the input query and the retrieved documents into feature representations—such as dense or sparse vectors—before they are fed into the model. By operating on feature representations rather than raw text, feature-level integration allows for more flexible manipulation of the input data.

5.6.2 Intermediate Layer Integration. Intermediate Layer Integration refers to integrate the external knowledge into the hidden layers of the generators. Attention-based Methods are one of the common approaches within Intermediate Layer Integration. A prominent example is the RETRO model[23], which introduces a novel cross-attention module to merge retrieved information with the model's intermediate representations. TOME [47] introduces a "Mention Memory"

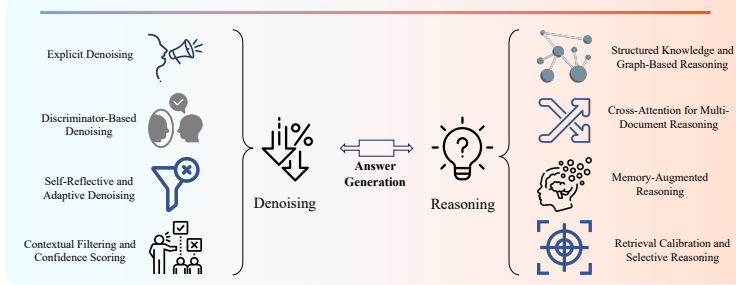


Fig. 6. Answer generation strategies utilized by RAG, including Denoising and Reasoning.

mechanism to integrate external knowledge into Transformers by storing and retrieving entity mention representations. The LongMem[47] framework uses an adaptive residual network for memory retrieval, which incorporates attention mechanisms to efficiently access and retrieve relevant long-term memory. Another approach[67] of intermediate layer integration is based on the lightweight (weighted) additions. These methods learn a weight for the embeddings of the top- k retrieved documents and then use weighted addition embeddings to incorporate them into the intermediate layers of the model.

5.6.3 Output Layer Integration. Output Layer Integration refers to integrating retrieved knowledge at the output layer of the generators. This approach typically combines the logits of the retrieved knowledge with the model's output logits, enabling enhanced generation. Output Layer Integration can be categorized into two main branches: ensemble-based integration and calibration-based integration. Ensemble-based integration aggregates logits from retrievals, such as in kNN-LM[113], where the probabilities of the nearest neighbors are interpolated with the model's predictions to improve generalization and robustness. Calibration-based integration, on the other hand, uses retrieval logits to refine the model's prediction confidence, as seen in confidence-enhanced kNN-MT[101]. For example, EDITSUM[134] enhances code summarization by incorporating prototype logits, and MA[65] applies a similar framework for image captioning.

5.7 Answer Generation

The generation component in RAG systems is central to producing responses that are both accurate and contextually relevant. However, achieving high-quality outputs in RAG models requires overcoming two major challenges: handling noisy retrievals and enabling effective reasoning over diverse information. This section examines these challenges by focusing on Denoising and Reasoning, each offering critical advancements in generation quality.

5.7.1 Denoising. Denoising is essential in RAG models to mitigate the influence of irrelevant, contradictory, or otherwise misleading information retrieved from large knowledge bases. Noise in retrievals can severely impact the factual accuracy and coherence of generated outputs, making denoising mechanisms indispensable in RAG pipelines.

Explicit Denoising Techniques. One effective approach to denoising is through explicit supervision. InstructRAG[241] introduces rationale generation, where the model is instructed to generate intermediate rationales that clarify

the relevance of each retrieved document. These rationales effectively filter noise by guiding the model to focus on more accurate and contextually relevant content. Similarly, the REFEED framework[271] refines initial answers by leveraging large language models to reassess the factual accuracy of the response based on retrieved data, iteratively discarding less relevant information.

Discriminator-Based Denoising. Another approach is using discriminative filtering mechanisms. For example, the COMBO framework[279] pairs generated paragraphs with retrieved ones using a pretrained discriminator, which assesses the coherence and relevance of each pair before the final generation phase. This discriminator-based approach ensure that irrelevant or contradictory information is identified and filtered, minimizing the risk of hallucinations.

Self-Reflective and Adaptive Denoising. Self-RAG[13] introduces a self-reflection mechanism, where the model critiques and revises its own responses by evaluating them for coherence and factuality. This method offers a dynamic way to handle noise, as the model can iteratively filter out untrustworthy or irrelevant information through self-evaluation. Additionally, adaptive retrieval strategies[100] allow the model to retrieve documents based on task-specific criteria, dynamically adjusting the retrieval scope and filtering intensity to optimize for relevance and quality, shown in Figure 6.

Contextual Filtering and Confidence Scoring. Recent methods have explored the use of confidence scoring to weigh the reliability of retrieved documents. For instance, models can assign scores to retrieved information based on its alignment with the query, filtering out low-confidence retrievals before integration[176]. This approach leverages confidence thresholds to systematically exclude noisy documents, making it especially useful for open-domain question answering where relevance varies significantly.

In summary, denoising methods—whether through rationale generation, discriminative filtering, self-reflection, or confidence scoring—play a critical role in ensuring that only the most relevant and accurate information is used in RAG models, thereby reducing hallucinations and enhancing overall output quality.

5.7.2 Reasoning. Beyond denoising, reasoning is essential for tasks that require synthesizing information across multiple documents. Effective reasoning enables the model to contextualize retrieved information, establish logical coherence, and generate responses that accurately reflect complex relationships.

Structured Knowledge and Graph-Based Reasoning. One advanced reasoning approach involves integrating structured knowledge sources, such as knowledge graphs, to assist in complex relational reasoning. Think-on-Graph 2.0[156] introduces a framework that combines knowledge graphs with unstructured text, allowing the model to reason over structured relationships between entities. By leveraging a graph structure, the model gains a deeper contextual understanding, improving its ability to answer queries involving complex relationships.

Cross-Attention for Multi-Document Reasoning. To facilitate reasoning across multiple documents, the RETRO model[23] employs chunked cross-attention, enabling the generative model to attend to relevant information across retrieved text chunks. This cross-attention mechanism helps maintain contextual coherence, particularly in open-domain settings where information spans several documents. Cross-attention has also been explored in augmented kNN-based approaches[113], where kNN attention layers allow the model to leverage neighborhood information during generation, providing a seamless integration of contextually relevant content.

Memory-Augmented Reasoning. Memory-augmented reasoning, as seen in models like EAE[67] and TOME[47], integrates entity-specific memory modules. These models store and dynamically retrieve information related to entities, allowing the generative model to maintain consistency and coherence over time. Memory modules are particularly

beneficial in tasks that require longitudinal consistency or multi-step reasoning, as they allow the model to recall entity-specific details across different parts of the conversation or document.

Retrieval Calibration and Selective Reasoning. Another critical advancement in reasoning is retrieval calibration, where models are trained to prioritize certain retrieved information based on its contextual relevance. Calibration techniques [105] help models identify the most crucial information, filtering out less relevant retrievals before reasoning takes place. Selective reasoning frameworks like those proposed by Wu et al. [15] enable the model to evaluate retrieved information in stages, ensuring that only the most contextually relevant data contributes to the final answer.

Hierarchical and Multi-Pass Reasoning. For complex queries requiring multi-step reasoning, hierarchical or multi-pass reasoning models [193] allow the model to process retrieved information iteratively, refining its understanding with each pass. Multi-pass reasoning is especially effective for tasks involving causal or temporal dependencies, as it enables the model to "revisit" previous knowledge in light of new information, creating a layered understanding that improves the accuracy and coherence of the response.

Overall, the combination of sophisticated denoising and reasoning techniques enhances the performance of RAG models. Denoising ensures that only the most relevant information reaches the generation phase, while reasoning structures and integrates this information coherently, yielding high-quality, contextually robust responses. Continued advancements in these areas are essential for building RAG systems that are reliable, efficient, and capable of handling complex, real-world queries.

5.8 Knowledge Citation

Citation in RAG is crucial for ensuring transparency, trustworthiness, and factual grounding of the model's responses. By attributing generated content to verifiable sources, it allows users to easily validate the information, reduces the burden of claim verification, and improves the evaluation process. Additionally, effective citation helps mitigate hallucinations, reinforcing the factual integrity of the model's outputs. In this subsection, we introduce strategies for generating citations in RAG and explore the evolving granularity of citations.

5.8.1 Strategies for Citation Generation. There are two main strategies for generating citations in language models: simultaneous citation generation and post-generation citation retrieval. Simultaneous generation, used by models like WebGPT[171], GopherCite[164], and RECLAIM[248], retrieves information in real time during response generation. This method ensures that the answer and citation are closely aligned, reducing hallucinations and improving factual accuracy. By contrast, post-generation citation, used by models like RARR[70] and LaMDA[229], involves generating the answer first and retrieving citations afterward. While this approach reduces computational complexity, it increases the risk of discrepancies between the response and the cited sources, as the answer is produced independently of the citations. Both approaches offer unique benefits: simultaneous generation provides stronger factual grounding, while post-generation citation offers greater flexibility in response generation.

5.8.2 Advances in Citation Granularity. Citation granularity—the level of detail provided in a citation—has improved significantly in recent models. Early models like LaMDA used coarse-grained references, often citing entire documents or URLs, which, while helpful for factual grounding, required users to sort through irrelevant information. More recent models, such as WebGPT, WebBrain [191] and GopherCite, have moved toward fine-grained citations. These models retrieve specific snippets of evidence or focus on comprehending long documents to support individual claims. The RECLAIM model exemplifies the highest level of citation granularity, linking individual claims to exact sentences in the source material. Similarly, longcite[276] proposed CoF (Coarse to Fine), a novel pipeline that leverages existing

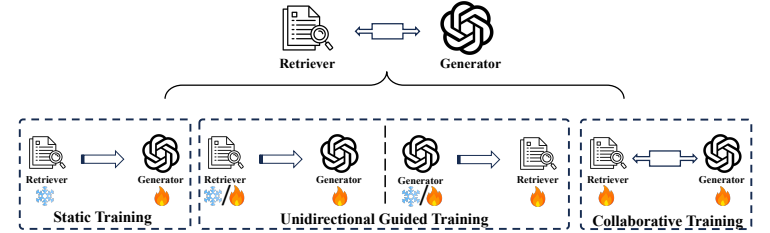


Fig. 7. RAG training includes static training, unidirectional guided training, and collaborative training.

LLMs to automatically generate precise sentence-level citations. Think&Cite formulates the generation of sentences with attribution as a multi-hop reasoning problem and solves it using Self-Guided Monte Carlo tree search[132]. This progression enhances the efficiency of fact-checking by reducing the user's effort to verify content and ensuring the information is easy to trace back to the original source.

6 ADVANCED RAG APPROACHES

Advanced RAG approaches, defined as methods beyond basic RAG models, encompass a range of cutting-edge techniques designed to overcome the limitations of basic RAG systems. These approaches aim to enhance the capabilities of RAG systems in multiple dimensions, including training optimization, multimodal processing, memory enhancing, and agentic reasoning. Specifically, this section delves into four key advancements: RAG Training, which focuses on improving the synergy between retrieval and generation; Multimodal RAG, which integrates multiple sensory modalities for enriched outputs; Memory RAG, which incorporates long-term memory to improve contextual reasoning and personalization; and Agentic RAG, which introduces iterative and dynamic optimization mechanisms to handle evolving information needs. Together, these approaches push the boundaries of what RAG systems can achieve, enabling them to address complex, dynamic, and specialized tasks.

6.1 RAG Training

Training RAG models requires balancing the optimization of both retrieval and generation components to achieve optimal performance. Effective training strategies ensure the retriever fetches relevant information while the generator produces coherent and accurate outputs. This section reviews various methods in RAG training, including static training, unidirectional guided training, and collaborative training, shown in Figure 8. Each approach offers distinct benefits and challenges, affecting the effectiveness and adaptability of RAG models across applications. By exploring these paradigms, we can enhance the integration of retrieval and generation processes, ultimately improving RAG performance.

6.1.1 Static Training. Static optimization training is a straightforward yet effective approach where either the retriever or the generator remains fixed during the training process, allowing optimization to focus solely on the other component. This method is particularly advantageous in scenarios with limited computational resources or when rapid deployment is essential. For instance, fixing the retriever while optimizing the generator allows the system to benefit from established

retrieval mechanisms, such as traditional BM25[204] or pre-trained models like BERT [49], without the overhead of simultaneously training the retriever. This can lead to faster training cycles and reduced resource consumption. However, the primary drawback of static optimization is the potential compromise in overall system performance. Since only one component is being optimized, the synergy between retrieval and generation may not be fully realized, potentially limiting the model’s ability to adapt to specific tasks or domains. To mitigate this, careful selection of the fixed component and the optimization process is essential to ensure that the evolving component can effectively leverage the fixed information.

6.1.2 Unidirectional Guided Training. Unidirectional guided training introduces a directional optimization strategy where one component directs the training of the other. This approach can be categorized into two distinct methodologies: retriever-guided generator training and generator-guided retriever training.

Retriever-Guided Generator Training. In retriever-guided generator training, the retriever plays a central role in shaping the training of the generator. By leveraging high-quality retrieved documents, the generator can be fine-tuned to better integrate and utilize external information. For example, RETRO [23] employs a pre-trained BERT model as the retriever to provide relevant context that enhances the generator’s output. Similarly, RALMs [264] use a pre-trained COLBERTV2 retriever to fine-tune large language models (LLMs), thereby improving the generator’s capacity to incorporate retrieved data effectively. ITER-RTGEN [217] utilizes S-BERT to guide the fine-tuning of a T5 generator, ensuring that the generated text aligns closely with the relevant retrieved information. Moreover, SMALLCAP [201] integrates CLIP as a retriever to direct a GPT-2 model in generating accurate and contextually appropriate captions. This guided approach ensures that the generator benefits from a steady stream of relevant information, enhancing the overall quality and relevance of the generated content.

Generator-Guided Retriever Training. Conversely, generator-guided retriever training focuses on optimizing the retriever based on the generator’s performance and requirements. In this paradigm, the generator’s ability to produce coherent and accurate text influences the retriever’s selection process. DKRR [95] leverages the generator’s attention scores to fine-tune the retriever, enhancing its capability to select the most pertinent information. AAR [272] employs smaller language models to generate supervision signals that guide the retriever’s training, ensuring that the retrieved documents are optimally aligned with the generator’s needs. RA-DIT [140] fine-tunes large language models before training the retriever, fostering better alignment and synergy between the two components. Additionally, UPRISE [33] uses a frozen LLM to guide the fine-tuning of a prompt retriever, thereby improving its effectiveness in retrieving data that the generator can utilize more effectively. This bidirectional influence ensures that the retriever evolves in tandem with the generator, fostering a more integrated and efficient RAG system.

6.1.3 Collaborative Training. Collaborative optimization training adopts a co-optimization strategy, simultaneously refining both the retriever and the generator to achieve optimal overall system performance. This holistic approach ensures that improvements in one component reinforce enhancements in the other, fostering a synergistic relationship that maximizes the capabilities of the entire RAG system. By jointly training both components, the system can better align the retrieval of relevant documents with the generation of accurate and coherent text. Notable implementations of this approach include RAG [126] and the work by [81], which utilize a joint training paradigm in conjunction with Maximum Inner Product Search (MIPS) [53, 200] to optimize the retrieval process effectively. This integrated training methodology allows the retriever to progressively improve its relevance scoring in response to the generator’s feedback, while the generator concurrently learns to better utilize the retrieved information. The result is a more robust and

adaptive RAG system capable of handling a wider range of tasks and adapting to varying input contexts with greater precision and fluency.

6.2 Multimodal RAG

Multimodal RAG extends traditional text-based RAG systems by incorporating multiple modalities including image, audio, video, etc. Compared to text-based RAG systems, multimodal RAG faces two main challenges. First, it needs to effectively represent and retrieve knowledge across different modalities, requiring sophisticated approaches to transform diverse data types into unified representations and enable cross-modal search. Second, after retrieving multimodal knowledge, the system must understand and utilize information across modalities to generate appropriate responses, demanding advanced techniques for multimodal understanding and generation. This section explores these two aspects: how multimodal RAG systems represent and retrieve diverse types of information, and how they understand and integrate this multimodal knowledge to produce contextually grounded responses.

6.2.1 Multimodal Representation and Retrieval. The foundation of multimodal RAG systems lies in their ability to effectively represent and retrieve information across different modalities. This requires both sophisticated embedding models to transform diverse data types into unified vector representations and specialized retrieval mechanisms to efficiently search across modalities. Various approaches have been developed to address these fundamental challenges:

Different modalities require specialized approaches for effective representation and retrieval. For visual content, models like NFNet[25] and Vision Transformer (ViT)[55] excel at extracting rich visual features, while CLIP[195] has revolutionized cross-modal retrieval by learning aligned representations of images and text through contrastive learning. In the audio domain, models like Wav2Vec 2.0[16] specialize in learning robust representations directly from raw waveforms, while CLAP[247] enables sophisticated audio search through audio-text alignment. Video content presents additional challenges due to its temporal nature, addressed by models like ViViT[9] and VideoPrism[281] that capture both spatial and temporal dynamics in compact feature vectors. These various approaches enable efficient cross-modal search and retrieval, supporting applications from visual question-answering to temporal video localization.

The effectiveness of multimodal RAG systems fundamentally depends on these representation and retrieval capabilities. While significant progress has been made with text embedding models like BGE[130] and NV-Embed[124], extending these capabilities to handle multimodal data remains challenging. Beyond the core modalities discussed, emerging data types such as code, structured data, and graph-structured information require specialized representation and retrieval approaches. The development of more sophisticated cross-modal retrieval techniques represents a crucial direction for advancing multimodal RAG systems.

6.2.2 Multimodal Understanding and Generation. Building upon effective multimodal representation and retrieval as the foundation, multimodal understanding and generation capabilities are essential for enhancing RAG systems beyond text-only processing. While representation and retrieval enable access to relevant multimodal content, the system must further process this information to comprehend cross-modal relationships and generate coherent outputs. Traditional RAG systems, primarily designed for textual data, often struggle to process and generate information across different modalities, leading to incomplete or inaccurate responses when multimodal content is crucial. Modern multimodal RAG systems address these limitations through sophisticated approaches to integrate understanding and generation across various modalities.

Models like CLIP[195] have pioneered the alignment of vision and language by embedding both data types into a shared space, enabling effective cross-modal retrieval. Building on this foundation, systems like MuRAG[32] enhance

Table 4. Comparison of Different Memory Types in RAG Systems

Aspects	Implicit Memory	Explicit Memory	Working Memory
Storage Form	model parameter	KV cache	plain text
Memory Term	long-term	long-term	short-term
Knowledge Scope	universal	domain-specific, personalized	task-specific, contextual
Usage Frequency	high	medium	low
Update Cost	high	low	low
Read Cost	low	low	high

question-answering by retrieving both images and text, while RA-CM3[262] extends this capability by enabling simultaneous retrieval and generation of text and images. Recent developments like Transfusion[286] demonstrate the potential of unifying language modeling with diffusion to train a single transformer that can both understand and generate across modalities. Similarly, Show-o[251] presents a unified transformer that combines autoregressive and discrete diffusion modeling to flexibly support a wide range of vision-language tasks including visual question-answering, text-to-image generation, and text-guided inpainting/extrapolation. Other advances like VisRAG[268] showcase the potential of multimodal RAG by leveraging both visual and textual information to achieve more comprehensive document understanding, while LA-RAG[135] advances speech processing by incorporating fine-grained token-level speech retrieval.

6.3 Memory RAG

Traditional RAG systems primarily rely on two extremes of knowledge storage and access: implicit memory embedded in language models through pre-training, and working memory that directly retrieves and processes raw text chunks. While language models excel at storing compressed general knowledge in their parameters, and traditional RAG effectively handles immediate context through text retrieval, there exists a crucial gap between these two approaches. Memory RAG bridges this gap by introducing explicit memory mechanisms that serve as an intermediate layer. This approach is particularly valuable for scenarios requiring comprehensive understanding of long documents or maintaining personalized knowledge over time, such as book comprehension and technical documentation analysis, where it can maintain hierarchical representations of document structure and key insights, as well as in personalization scenarios where it efficiently stores and updates compressed representations of user preferences and behavior patterns.

The value of this intermediate approach lies in its unique advantages over both extremes. Unlike the rigid implicit memory in language models that requires costly retraining to update, and unlike the computationally intensive working memory that processes raw text directly, explicit memory in Memory RAG provides a flexible and efficient middle ground. It enables compressed, structured representations of knowledge that are both more updateable than model parameters and more efficient than raw text processing. By implementing these distinct memory mechanisms, Memory RAG creates a more complete knowledge processing pipeline that combines the strengths of all three memory types. This section first explores these different types of memory and their complementary roles, then demonstrates how their technical implementation contributes to more efficient and scalable information processing in RAG systems.

6.3.1 Types of Memory. Implicit Memory.

Implicit memory, analogous to human implicit memory [18, 40, 69], represents knowledge that is deeply embedded but not consciously accessible. Just as humans develop unconscious skills like riding a bicycle or typing without

conscious thought, RAG systems encode implicit knowledge within their model parameters through training. This form of memory manifests through the pre-trained weights of both retriever and generator components, storing global knowledge and domain-specific patterns that are frequently accessed during inference. While implicit memory enables fast inference and provides a foundation for general knowledge, it shares limitations with its human counterpart: it's costly to acquire, difficult to modify, and not directly interpretable. Updates to this type of memory typically require model retraining.

Explicit Memory. Explicit memory parallels human explicit memory [182] - the conscious recollection of facts and experiences. In RAG systems, it serves as a compressed, structured representation of long-term knowledge that captures high-level understanding of information sources. Unlike simple document retrieval, explicit memory stores abstract representations such as comprehensive understanding of entire books, user behavior patterns over time, or structured summaries of knowledge bases. This intermediate form between raw text and model parameters enables efficient storage while maintaining semantic richness. It operates at a medium access frequency, supporting stable but updatable knowledge representation. The key advantage of explicit memory lies in its ability to maintain holistic understanding that cannot be achieved through simple passage retrieval, while still remaining more flexible than implicit memory encoded in model parameters.

Working Memory. Working memory mirrors human short-term memory [42, 43], serving as a temporary storage for immediate processing. In RAG systems, this manifests as the retrieved plain text chunks that are incorporated into the prompt context window, along with recent conversation history and intermediate computation results. Just as humans maintain recent information in mind while performing complex tasks, RAG's working memory temporarily holds relevant retrieved passages to inform the current generation task. This type of memory is characterized by high-frequency access, rapid updates with each new task, and temporary storage. It faces similar capacity constraints to human working memory - it's limited by computational resources and context window size, just as humans can only hold a finite amount of information in immediate awareness. The retrieved passages in the prompt serve as the temporary workspace where the model processes and synthesizes information to generate appropriate responses.

6.3.2 Technical Implementation. Memory RAG primarily focuses on explicit memory, which takes the form of sparse key-value caches that serve as an intermediate representation between raw text and model parameters. Memory RAG converts knowledge from raw text into more compact and structured explicit memories through key-value caching, uses these memories to guide and optimize the retrieval process, and reduces dependence on real-time retrieval to improve system efficiency. It reduces computational costs through efficient memory management, enhances long-text processing capabilities by storing and selectively retrieving information, and improves answer accuracy and interpretability through structured knowledge representation.

Memory³ [258] employs a progressive compression mechanism that transforms raw input tokens into explicit memories through a two-stage pre-training process. The first stage, called warmup stage, focuses on memory formation without explicit memory to develop the model's basic reading comprehension abilities. The second stage, called continual train stage, introduces explicit memories and enhances the model's ability to utilize these memories during generation. This approach particularly emphasizes memory sparsification and storage optimization through sparse key-value storage, with several innovative designs: it selects only the first half of attention layers as memory layers, reduces key-value heads through grouped query attention, and employs token-level sparsification to select only the most relevant tokens for each head. These optimizations enable efficient handling of large-scale knowledge bases while significantly reducing storage requirements.

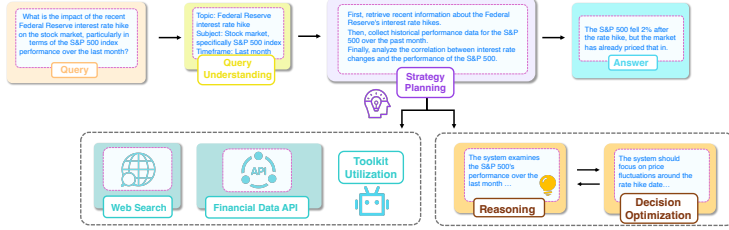


Fig. 8. Agentic RAG consists of query understanding & strategy planning, toolkit utilization, reasoning & decision optimization.

MemoRAG [91] takes a unique approach by employing a lightweight but long-range LLM as a global memory system. This design adopts a dual-system architecture: a lightweight LLM serves as the memory module to form global memory of the database, while a more expressive LLM acts as the generation module. To form the global memory, MemoRAG introduces a progressive memory formation process where the memory module processes the input sequence through multiple attention layers. During this process, it maintains a KV cache of previous memory tokens and employs special weight matrices for memory formation, allowing the model to transform raw input tokens into compressed memory representations. The memory tokens are updated through attentive interactions with both the input context and previously formed memories, enabling the model to capture and preserve essential information across the entire database. The memory module first generates draft answers that serve as retrieval clues, effectively reducing the search space and improving retrieval precision. This approach is particularly effective for handling ambiguous information needs, distributed evidence gathering, and information aggregation tasks.

CAG (Cache-Augmented Generation) [26] adopts a more radical approach by completely eliminating real-time retrieval from its workflow. Instead of performing retrieval during generation, it pre-computes key-value caches for all relevant documents and uses these caches directly during the generation process. The system operates in three phases: external knowledge preloading, where all relevant documents are preprocessed and formatted into a KV cache; inference, where the precomputed cache is used directly for generation without retrieval; and cache reset, which efficiently maintains system performance across multiple conversation turns. This design is particularly effective for scenarios with well-defined, manageable knowledge bases, where the complete set of relevant information can be pre-processed and stored in cache. The direct cache access strategy not only improves response time but also ensures stable performance in multi-turn interactions, while significantly reducing system complexity by removing the need to integrate separate retrieval and generation components.

6.4 Agentic RAG

Agentic RAG is an advanced framework that integrates autonomous agents with RAG technology, significantly enhancing the performance of the information retrieval and generation processes. This strategy introduces agent-based decision-making to dynamically manage retrieval strategies, improving complex problem-solving, knowledge management, and

generation processes. In Agentic RAG, autonomous agents are responsible for query understanding, tool utilization, and reasoning optimization.

6.4.1 Query Understanding & Strategy Planning. Query understanding and strategy planning is a critical step in Agentic RAG, enabling the effective understanding of user queries and the formulation of appropriate retrieval strategies. Agents analyze query complexity and topics to determine the priority and methods of retrieval. For instance, AT-RAG[202] improves retrieval efficiency and reasoning accuracy for complex multi-hop queries through topic filtering and iterative reasoning. Additionally, REAPER [3] introduces a reasoning-based retrieval planner, enhancing performance in complex queries, which demonstrate how query understanding can provide more efficient retrieval pathways for Agentic RAG.

6.4.2 Toolkit Utilization. A key feature of Agentic RAG is its ability to utilize multiple tools. Agents are capable of employing traditional search engines, calculators, APIs, and other external tools to enhance retrieval and reasoning capabilities. For example, AT-RAG [202] integrates various tools to expand the scope of retrieval, while RAGENTIC[2] leverages multi-agent collaboration and tool usage to tackle complex tasks. This integration of multiple tools allows Agentic RAG to adapt flexibly to different task requirements, thus improving the system's efficiency and adaptability.

6.4.3 Reasoning & Decision Optimization. Reasoning and decision optimization is a core mechanism in Agentic RAG systems, ensuring that agents make optimal decisions based on constantly changing information and environments. Through reasoning, agents can evaluate the reliability of multiple information sources, perform multi-step reasoning, and optimize retrieval strategies based on reasoning outcomes. PlanRAG [1] improves the performance of generative large language models in decision-making by employing a plan-then-retrieve approach. Moreover, REAPER [3] excels in decision optimization by enhancing retrieval planning based on reasoning, thereby improving the response speed of RAG systems. These decision optimization methods not only enhance retrieval accuracy but also strengthen the system's ability to handle complex tasks.

7 COMPREHENSIVE EVALUATION STRATEGIES FOR RAG

Evaluating RAG systems requires assessing both their effectiveness and efficiency. Effectiveness evaluations examine how well a model retrieves relevant context, integrates it into coherent answers, and maintains overall accuracy. Efficiency assessments consider the computational resources, response latency, and scalability of the retrieval and generation processes.

7.1 Assessing the Effectiveness of RAG

Evaluating the effectiveness of RAG is key in determining their capability to harness external knowledge for generating reliable responses. The following subsections delve into specific metrics designed to evaluate key aspects of RAG effectiveness, including query-context relevance, context-answer coherence, and query-answer accuracy.

7.1.1 Assessing Query-Context Relevance. In RAG systems, the ability to retrieve pertinent information in response to user queries is fundamental to providing large language models (LLMs) with the necessary context for accurate reasoning and generation [36]. This process involves efficiently integrating data from diverse knowledge sources and implementing robust filtering mechanisms to extract relevant information while minimizing noise from less reliable or extraneous data. To evaluate the effectiveness of RAG systems in this regard, several benchmarks and metrics have been developed. Multiple frameworks including ARES[208], RAGAS[61], and TruLens[233] employ Context Relevance metrics to assess how well the retrieved context aligns with and supports the user's query. Additionally, the KILT[188]

benchmark, designed for knowledge-intensive tasks, offers metrics such as R-precision and Recall@k to evaluate the quality of the retrieved context, emphasizing its relevance and comprehensiveness concerning the original query. While these benchmarks share a common goal of ensuring that the retrieved context meaningfully supports the user’s query, they differ in their specific approaches and emphases. ARES, RAGAS, and TruLens focus on the immediate alignment between query and context, whereas KILT provides a broader evaluation by considering the richness and completeness of the retrieved information. Collectively, these metrics are critical for assessing a RAG system’s ability to retrieve and provide high-quality, relevant information, enabling LLMs to generate accurate and contextually grounded responses.

7.1.2 Evaluating Context-Answer Coherence. The relationship between retrieved context and generated answers is crucial for ensuring response quality in RAG systems. The generation component must produce responses that are both coherent and faithful to the retrieved context, thereby minimizing hallucinations or unsupported claims. To assess this capability, various benchmarks and metrics have been developed. For instance, CRUD [155] employs Recall and Precision metrics to evaluate the accuracy of generated answers, focusing on the correctness and completeness of the responses. ARES and RAGAS introduce the Answer Faithfulness metric, which measures the extent to which generated answers remain loyal to the retrieved passages, ensuring that the content is free from hallucinations. RECALL [147] proposes two key metrics: Accuracy, assessing the factual correctness of the generated answer, and Misleading Rate (M-Rate), evaluating the frequency of misleading information produced. Additionally, TruLens introduces the concept of Groundedness, which involves breaking down the LLM’s generated answer into individual claims and verifying each against the retrieved context, ensuring that the output is supported by concrete evidence. While these benchmarks share a common goal of evaluating the fidelity and coherence of generated answers, they differ in their specific approaches. CRUD focuses on the accuracy metrics of Recall and Precision; ARES and RAGAS emphasize faithfulness to the retrieved context; RECALL assesses both accuracy and the propensity to produce misleading information; and TruLens delves into the granular verification of individual claims.

In addition to these core metrics, recent work has focused on enhancing the reliability of RAG systems through high-quality citation and accurate attribution[288]. For instance, LongBench-Cite[276] is designed to evaluate models in Long-Context Question Answering with Citations (LQAC), where the relevance and accuracy of citations are essential for maintaining transparency. The RECLAIM framework[248] further develops this idea with the Correct Attribution Score (CAS), assessing whether generated answers are fully supported by citations, and the Citation Redundancy Score (CRS), which minimizes unnecessary citations to enhance response clarity. Metrics such as Verifiability and Consistency Ratio (CR) in RECLAIM ensure that the retrieved data genuinely supports the generated content.

Moreover, RARR [70] introduces sentence-level AIS and Levenshtein distance to measure citation accuracy, ensuring that references retain their original meaning. WebGPT [171] adds a layer of reliability by distinguishing between imitative and non-imitative errors to improve error correction. Additionally, ALCE [72] uses citation recall and precision to evaluate citation relevance and conciseness. Collectively, these metrics contribute to a comprehensive framework for evaluating RAG’s ability to generate contextually grounded, accurate, and transparently attributed responses, emphasizing reliability and trustworthiness in practical applications.

7.1.3 Measuring Query-Answer Accuracy. In RAG systems, accurately measuring the alignment between user queries and generated answers is essential for assessing overall performance. Traditional question-answering (QA) evaluations often rely on accuracy metrics, which focus on whether the system provides the correct answer. However, RAG systems introduce additional complexities, requiring more nuanced evaluation approaches. A significant challenge in RAG evaluation is addressing hallucinations—instances where the model generates content not grounded in the

retrieved context. To tackle this, benchmarks such as CRAG[260] incorporate metrics like hallucination rate and missing rate alongside traditional accuracy, calculating the difference between accuracy and hallucination rate to penalize hallucinations, thereby prioritizing factually correct responses. Other frameworks, such as RAGAS, ARES, and TruLens, include an Answer Relevance metric to evaluate how well the generated response aligns with the original query. Similarly, CRUD introduces RAGQuestEval to assess answer relevance and correctness, while the KILT benchmark evaluates RAG systems based on specific downstream tasks, offering a task-oriented perspective on generated answers. Additionally, the RGB framework[27] extends evaluation criteria by introducing rejection rate, error detection rate, and error correction rate, which test the system’s ability to identify and reject incorrect or misleading information. Collectively, these approaches underscore a shift toward comprehensive RAG evaluation, moving beyond simple accuracy to include relevance, factual accuracy, and error-handling capabilities. This shift highlights the growing emphasis on building RAG systems that are not only effective but also reliable and trustworthy for real-world applications.

7.2 Evaluating the Efficiency of RAG

The efficiency of RAG systems is crucial for their successful deployment in production environments, where low latency and high throughput are essential for delivering responsive services. To evaluate RAG efficiency, several key performance metrics need to be measured, including latency, throughput, and resource utilization. Latency refers to the end-to-end response time, encompassing both the retrieval and generation phases. Efficient retrieval mechanisms—such as dense or sparse retrievers—play a significant role in reducing latency; however, dense retrievers often demand higher computational resources, which can increase processing time. Throughput, which indicates how many queries the system can process per second, is vital for high-traffic environments, where scaling capabilities are tested. Additionally, resource utilization, including CPU/GPU usage and memory consumption, reflects how effectively the system makes use of hardware resources without introducing unnecessary overhead.

The RAG-Performance library[214] is designed to simulate high-volume data environments, pushing RAG models to their limits in terms of response time and throughput. By benchmarking different retrieval and generation mechanisms, it provides a comprehensive view of how different configurations affect system performance. Other research also considers not only execution time but also hardware utilization, offering insights into the trade-offs between effectiveness and efficiency[66, 106, 209]. To fully understand the operational efficiency of RAG systems, it is essential to benchmark these metrics under various configurations and scalability tests. This ensures that the system remains responsive and cost-effective, even under varying loads. Future research directions may focus on developing more comprehensive efficiency evaluation frameworks that consider both computational resources and system responsiveness in real-world deployment scenarios.

8 DOWNSTREAM TASKS AND APPLICATIONS

In this section, we will comprehensively summarize downstream tasks and domain-specific applications of RAG.

8.1 Downstream Tasks

Generally, downstream tasks cover diverse areas, including question answering, information extraction, text generation, and classification. Table 5 summarizes each task type and their common sub-tasks, highlighting areas where RAG models and similar architectures are frequently applied.

In Question Answering (QA), RAG models demonstrate significant advantages by retrieving targeted information that supports accurate and relevant answers:

Task	Sub Task	Dataset
Question Answering (QA)	Single-hop QA	Natural Question (NQ) [119]
		TriviaQA (TQA) [108]
		SQuAD [198]
		Web Questions (WebQ) [20]
		PopQA [162]
		CRAG [259]
	Multi-hop QA	HotpotQA [261]
		2WikiMultiHopQA [89]
		StrategyQA [75]
		MuSiQue [231]
		MultiHop-RAG [227]
Information Extraction	Entity Linking	ELI5 [62]
		WebCPM [192]
	Relation Extraction	NarrativeQA (NQA) [118]
Text Understanding	Text Classification	ZESHEL [150]
		CoNLL [90]
	Text Summarization	T-REx [59]
	Text Generation	ZsRE [125]

Table 5. Datasets for Downstream Tasks in RAG

- **Single-hop QA:** For straightforward factual questions, RAG retrieves a single document or passage that directly addresses the query, allowing the model to generate precise answers with minimal reasoning complexity.
- **Multi-hop QA:** The retrieval mechanism of RAG shines in multi-hop scenarios, where it must gather and synthesize information from multiple sources to answer questions that require connecting different facts or viewpoints. This multi-step retrieval capability enables RAG models to handle more sophisticated questions and provide in-depth responses.
- **Long-form QA:** In tasks requiring detailed, paragraph-length answers, RAG retrieves and combines multiple relevant documents, synthesizing a comprehensive and coherent response. This is particularly valuable for open-ended or complex questions, as it enables the model to produce nuanced and well-informed answers.

In Information Extraction, the ability of RAG to pull contextually relevant data from external sources enhances the accuracy and depth of entity and relation identification:

- **Entity Linking:** By accessing real-time information on specific entities, RAG models can more accurately link entities mentioned in text to standardized entries in a knowledge base, improving disambiguation and contextual understanding.

- **Relation Extraction:** The retrieval capability of RAG allows the model to gather background information on related entities, facilitating the accurate identification of relationships within the text. This supports the construction of structured knowledge, useful for applications like knowledge graph development.

In Text Understanding and Generation, RAG provides contextual depth by enabling the retrieval of supplementary information that enhances interpretability and relevance:

- **Text Classification:** For classification tasks, especially in specialized domains, RAG can retrieve background information or examples that support accurate categorization, improving the model performance in classification scenarios.
- **Text Summarization:** RAG can retrieve summaries or additional context from related documents, helping the model focus on the most relevant points for concise, informative summaries.
- **Text Generation:** In generative tasks, the retrieval capability of RAG allows it to incorporate real-time data, domain-specific knowledge, or additional context into generated content.

8.2 Applications of RAG

RAG models have broad applications across various fields, where they combine the strengths of information retrieval with language generation. This part explores key application areas where RAG models significantly enhance performance by providing accurate, contextually relevant, and real-time information retrieval capabilities.

8.2.1 AI for Science. In scientific research, RAG models provide powerful support by enabling researchers to access and integrate up-to-date information from extensive scientific databases, which enhances productivity and accelerates discovery. For instance, in materials science, large language models like GPT-4, combined with retrieval capabilities, can autonomously assist in complex tasks such as material design and data analysis, reshaping the research paradigm in this interdisciplinary field [267]. In chemistry, RAG models improve the handling of discipline-specific queries, supporting researchers in synthesis planning, reaction prediction, and understanding molecular interactions, which require precise scientific information beyond pre-trained knowledge[243]. In physics, RAG models enhance question-answering and content comprehension by retrieving relevant scientific context, allowing researchers and students to engage more deeply with complex physical concepts, thereby supporting more advanced scientific inquiry and problem-solving [5]. In the life sciences domain, BioRAG [237] functions as a specialized question-answering system, designed to support researchers in biology and related fields by providing efficient access to relevant scientific information. These examples underscore the versatility of RAG models in scientific fields, where they improve the retrieval and synthesis of domain-specific information.

8.2.2 Finance. In the financial domain, RAG models play a crucial role in enhancing the accuracy and relevance of information derived from complex data, supporting tasks like information extraction, question answering, and sentiment analysis. In question answering on financial documents, RAG models enhance response reliability by refining retrieval methods. Techniques like improved chunking and query expansion ensure that the model retrieves the most relevant information, resulting in more accurate and contextually appropriate answers in financial QA systems [216]. For financial sentiment analysis, RAG models offer substantial advantages by retrieving additional context, allowing large language models to make more accurate sentiment predictions. This capability aligns model outputs with the nuanced requirements of financial sentiment, supporting better-informed valuation and investment decisions [274]. Finally, RAG-augmented models provide valuable business insights by extracting data from various document types, such as

invoices and reports. This approach offers organizations a scalable and cost-effective solution for comprehensive data analysis and insight generation [10, 11]. These applications demonstrate how RAG models address specific challenges in finance, improving information retrieval and supporting more informed decision-making across the industry.

8.2.3 Education. In education and e-learning, RAG models enhance learning experiences by providing personalized, context-specific support. Intelligent tutoring systems using RAG overcome traditional limitations by delivering accurate, interactive assistance that improves student engagement and academic performance [168, 230]. Additionally, HiTA is a RAG-based platform that supports educators by positioning them within AI-assisted learning loops, functioning as a teaching assistant that increases course-specific learning satisfaction [141]. Furthermore, Education-Specific RAG AI (ES-RAG AI) ensures transparency and accountability in AI-generated educational content, aligning it with educational needs [58]. Overall, RAG models create adaptive learning environments, tailor content based on student progress, and assist educators in efficient curriculum development, enhancing both learning and teaching processes.

8.2.4 Healthcare. In healthcare, RAG has emerged as a transformative tool, enhancing how medical professionals access and utilize medical knowledge. By combining advanced retrieval mechanisms with generative capabilities, RAG significantly supports knowledge retrieval, decision-making processes, and personalized healthcare services. RAG applications span diverse areas, such as Traditional Chinese Medicine, where it retrieves specialized knowledge from texts like the Compendium of Materia Medica, aiding modernization efforts [146]. In infectious disease management, RAG-based chatbots provide timely, context-aware information to address healthcare resource shortages [197]. In precision medicine, RAG integrates NLP and genomic data to deliver personalized treatment recommendations for conditions like Multiple Myeloma [194]. These examples illustrate RAG’s versatility in healthcare, supporting both traditional and modern medicine, improving disease management, and advancing precision medicine.

8.2.5 Legal. In the legal sector, RAG models are instrumental in navigating complex legal texts, supporting legal research, document drafting, and client consultations. For example, CBR-RAG combines Case-Based Reasoning with RAG to structure retrieval processes, enhancing legal question-answering quality by ensuring contextually relevant cases inform the LLM’s outputs [245]. LegalBench-RAG introduces a benchmark emphasizing precise retrieval within the legal domain, focusing on extracting highly relevant text to support accurate analysis and mitigate context window limitations [189]. Additionally, HyPA-RAG, a hybrid parameter-adaptive system, addresses challenges such as retrieval errors and outdated information by dynamically adjusting retrieval parameters, significantly improving response accuracy in high-stakes applications like policy interpretation [109]. These implementations highlight how RAG models enhance efficiency, accuracy, and contextual relevance in legal research and documentation, ultimately enabling faster, more reliable legal services.

8.2.6 Industry. In industry, RAG models play a vital role in improving operational efficiency, real-time troubleshooting, and decision-making in complex scenarios by integrating retrieval with generative capabilities. For instance, RAG systems in industrial troubleshooting retrieve information from technical manuals, expert databases, and real-time sensor data, synthesizing it to offer context-specific solutions, thereby minimizing downtime and enhancing resolution rates [173]. In industrial maintenance, STMicroelectronics used RAG as cognitive assistants to combine technical document retrieval with generative capabilities to support advanced maintenance strategies, reducing errors and providing context-rich assistance [160]. Additionally, Intelligent Driving Assistance Systems use RAG models to provide real-time, context-based guidance from car manuals, significantly enhancing the driving experience [88, 117, 142].

Overall, RAG models bridge the gap between data retrieval and contextual generation in industrial applications, delivering timely insights that reduce operational challenges and improve efficiency across various sectors.

9 PROSPECTS AND FUTURE DIRECTIONS

9.1 GraphRAG

Significance. Along with the growing attention on knowledge-graph-based methods in retrieval-augmented generation (RAG) systems, effectively leveraging structured relationships for tighter interaction between text generation and retrieval has become a new research focus. By coupling retrieved external knowledge with graph structures, GraphRAG can establish explicit connections among different entities and their relations, leading to more interpretable and robust performance, especially for complex factual or reasoning tasks. In practice, GraphRAG can also be viewed as a form of context extension in traditional RAG, where nodes and edges in the graph serve as composable contexts for multi-hop reasoning and entity alignment.

Open issues. Although some studies have explored the integration of knowledge graphs into RAG, there remains a lack of systematic solutions to incorporate multi-source heterogeneous graph data (e.g., cross-domain or cross-lingual scenarios). Moreover, dynamically updating or correcting inaccurate or outdated graph structures during inference remains a major challenge. As the range of application scenarios grows, real-time responsiveness on large-scale graphs, efficient retrieval, and scalability in high-concurrency settings are becoming key directions for further exploration in GraphRAG research.

9.2 Multimodal RAG

Significance. Multimodal RAG is becoming increasingly prevalent, as real-world information is not limited to text but also includes images, videos, audio, and other sensor data. Effectively incorporating these heterogeneous modalities into RAG can significantly enhance reasoning capabilities, enabling more comprehensive understanding and generation in applications such as human-machine interaction, multimodal content creation, and medical diagnostics. By integrating visual, auditory, and textual information, Multimodal RAG can improve tasks like product recommendation, scene understanding, and multimodal QA, making retrieval-enhanced reasoning more robust and context-aware.

Open Issues. A major challenge in Multimodal RAG lies in how to effectively represent and retrieve information across different modalities, ensuring seamless fusion with large language models for reasoning and generation. Heterogeneous data sources often have distinct structures, resolutions, and levels of noise, requiring advanced alignment strategies to maintain consistency and interpretability. Moreover, real-time retrieval and reasoning over high-dimensional multimodal data demand efficient indexing, storage, and processing techniques. Future research should focus on scalable multimodal representation learning, cross-modal retrieval optimization, and adaptive fusion strategies, while also addressing issues such as privacy, robustness, and incremental updates in dynamic environments.

9.3 Personalized RAG

Significance. Personalized RAG is crucial for enhancing user experience in recommender systems, dialogue agents, and QA applications. By integrating user-specific knowledge, such as preferences and historical interactions, RAG can generate more relevant and context-aware responses. A practical and cost-effective approach to achieving personalized RAG is through memory-based mechanisms, where user-related knowledge is systematically extracted, stored, and retrieved. This enables the model to adapt dynamically to individual needs while maintaining efficiency and scalability.

Open Issues. The key challenges in Personalized RAG revolve around memory extraction, storage, knowledge updating, and fusion. Efficiently structuring and managing memory to ensure relevance and usability over time remains an open problem. Furthermore, updating personalized knowledge while mitigating forgetting or inconsistencies requires advanced strategies for lifelong learning. Privacy and security concerns also persist, as storing and utilizing personal information must comply with data protection regulations and prevent unauthorized access. Additionally, expanding Personalized RAG to multilingual and cross-domain scenarios requires further research to ensure adaptability across diverse linguistic and cultural contexts.

9.4 Agentic RAG

Significance. Agentic RAG operates like an intelligent investigator—it not only retrieves relevant documents but also analyzes their content, cross-references information, and simulates interviews to gain deeper insights. This paradigm introduces AI agents as intermediaries between users and knowledge sources, substantially enhancing retrieval-augmented reasoning. These agents can: interpret user queries, plan retrieval strategies, leverage diverse tools (e.g., vector search engines, web search, calculators, and APIs), and dynamically adjust their reasoning to optimize information retrieval and synthesis. Recent advancements, such as OpenAI’s DeepResearch² and Google’s research on intelligent retrieval systems³, highlight the potential of Agentic RAG. DeepResearch, in particular, exemplifies how agents autonomously decide when and how to search, engage in multi-round reasoning and retrieval cycles, and progressively analyze information to generate comprehensive research reports. By integrating agentic decision-making, RAG systems achieve greater autonomy, enabling them to handle complex, multi-step tasks beyond the reach of traditional RAG frameworks.

Open Issues. Despite its promise, Agentic RAG encounters notable challenges in structured multi-step reasoning, including maintaining interpretability, mitigating error propagation, and coordinating diverse knowledge sources effectively. While systems like DeepResearch[?] excel at identifying task-relevant and high-quality data sources, they are currently constrained to publicly accessible information, potentially overlooking insights from closed or proprietary datasets. Another challenge lies in the depth of analysis—current models tend to perform better at summarization than providing nuanced analytical insights. Developing reliable agents that can flexibly plan retrieval strategies while ensuring efficiency and correctness remains an open research area. Moreover, optimizing these agents for distributed and hardware-heterogeneous environments is critical for broader real-world deployment. Future work should focus on enhancing autonomous decision-making, integrating more robust reasoning mechanisms, and improving adaptive learning capabilities. Collaborating with domain experts to facilitate deeper analytical insights and expanding the scope of accessible data sources are essential directions for further advancing the effectiveness of Agentic RAG.

9.5 RAG and Generative Models

Significance. The integration of RAG with various generative models, including diffusion models and other deep generative architectures, opens up new possibilities for solving complex tasks. Beyond the synergy with large language models like ChatGPT or GPT-4, the incorporation of generative paradigms such as diffusion models can enhance both the diversity and fidelity of generated content. This broader integration extends RAG’s impact beyond traditional QA and dialogue tasks, fostering innovative applications in cross-domain scenarios such as scientific literature mining, medical imaging synthesis, and multimodal content generation.

²<https://openai.com/index/introducing-deep-research/>

³<https://research.google/blog/accelerating-scientific-breakthroughs-with-an-ai-co-scientist/>

Open Issues. Despite its potential, the combination of RAG with generative models remains challenging due to the inherent opacity of both retrieval-based and generative processes. The limited interpretability of interactions between external knowledge retrieval and generative components complicates model transparency and control. A more unified approach in designing architectures, training paradigms, and retrieval-augmentation strategies is essential for effectively leveraging these capabilities. Furthermore, robust mechanisms for uncertainty quantification, error correction, and adaptive retrieval are critical to mitigating inaccuracies and ensuring the reliability of generated content in real-world applications.

9.6 EdgeRAG

Significance. As edge computing gains prominence in applications such as IoT, mobile systems, and smart environments, the role of RAG in edge scenarios is becoming increasingly important. Deploying retrieval and generation capabilities at the edge, rather than relying solely on cloud-based processing, offers key benefits such as lower latency, reduced bandwidth usage, and enhanced local privacy protection. These advantages are particularly crucial for real-time applications and data-sensitive scenarios where offloading to the cloud is impractical or undesirable.

Open Issues. The resource constraints in edge environments, including limited computational power, storage, and caching capacity, pose significant challenges for efficient RAG deployment. Optimizing model compression, retrieval efficiency, and adaptive caching strategies is essential to balance performance and resource consumption. Additionally, maintaining reliability and consistency in RAG outputs under network instability or intermittent cloud connectivity remains an open problem. Given the security vulnerabilities of edge environments, robust defense mechanisms against adversarial attacks and unauthorized access are also critical for ensuring safe and trustworthy EdgeRAG applications.

9.7 Trustworthy RAG

Significance. As RAG systems are increasingly deployed in real-world applications, their interpretability and trustworthiness have become critical concerns. Trustworthy RAG emphasizes not only transparency in retrieval and generation processes but also the explainability of its underlying mechanisms and cited sources. Ensuring that both retrieval pathways and generated outputs are interpretable is essential for validating information accuracy, enhancing user trust, and supporting decision-making in high-stakes domains such as healthcare, finance, and legal analysis.

Open Issues. Despite growing attention to trust and explainability, there is no universally accepted framework for assessing the reliability of RAG outputs. Existing mechanisms struggle to fully address challenges such as unverifiable citations, hallucinations, and inconsistencies between retrieved knowledge and generated content. Moreover, concerns related to model bias, privacy, and regulatory compliance remain unresolved. Future work should focus on developing interpretable RAG architectures, rigorous citation verification techniques, and robust auditing mechanisms, ensuring alignment with emerging ethical standards and regulatory frameworks.

9.8 Developments of Benchmark and Evaluation

Significance. With the rapid evolution of RAG techniques, there is a growing need for robust evaluation methodologies and standardized benchmark datasets. Comprehensive, reproducible, and challenging benchmarks are crucial for making objective comparisons among different approaches and for identifying each method’s strengths and weaknesses. In real-world settings, where multimodal inputs, personalization, and online scenarios are increasingly common, it is necessary to balance and expand metrics encompassing accuracy, efficiency, robustness, and user experience.

Open issues. Existing benchmarks often focus on single-modality or offline use cases, making them insufficient to capture the diverse application requirements of future RAG deployments. Moreover, there is still no unified standard for quantifying model interpretability, trustworthiness, or uncertainty handling. As online services and privacy compliance become prevalent, evaluation methodologies also need to address real-time constraints and security considerations to provide more comprehensive experimental foundations for new RAG research and applications.

10 CONCLUSION

In this paper, we have conducted a systematic and extensive review of the most notable works to date on retrieval-augmented generation (RAG) from a knowledge-centric perspective. We proposed an overarching framework that organizes existing research into core modules—ranging from knowledge sourcing and embedding to retrieval and final answer generation—to reduce ambiguities in this rapidly evolving field. We thoroughly analyzed the associated challenges, particularly in knowledge parsing, integration, and context adaptation, and offered a classification scheme that includes emerging approaches such as multimodal and memory-augmented RAG. We also identified critical open issues and promising research directions, underscoring RAG’s potential to transform knowledge-intensive applications across diverse domains. It is our hope that this survey will provide readers with a solid understanding of the key components, main challenges, and ongoing developments in RAG, and shed light on new opportunities in future studies.

ACKNOWLEDGMENTS

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Received 20 February 2025